

Introduction to Renormalization

Polology, and field and mass renormalization

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Non-perturbative foundations

Introduction to Part I

A scattering experiment speaks the language of particles. It tells us which stable particles exist, what their masses and spins are, and with what amplitudes an incoming collection of particles can turn into an outgoing one. A quantum field theory is written in a different language: local operators, correlation functions and an algebra of observables. The first task is to connect the two.

That connection is already the beginning of renormalization. A mass parameter written in a Lagrangian is not automatically the mass measured in an experiment. Nor does a local field come with a preferred normalization: multiplying it by a constant changes its correlation functions without changing the particle it can create. Scattering theory tells us how to remove these ambiguities. The mass measured by the asymptotic particle is the physical mass, and a local field is physically normalized when its one-particle matrix element has the standard normalization used for the asymptotic state. Once the external particles have been normalized in this way, the scattering matrix, or S -matrix, supplies the physical information with which interaction strengths may be defined.

None of this can be justified by perturbation theory alone. The essential bridge is provided by non-perturbative theorems. Haag–Ruelle theory constructs asymptotic particles from local operators. The systematic study of pole singularities—which we shall call *polology*—shows that the same particle appears as an isolated pole of exact correlation functions. Lehmann–Symanzik–Zimmermann (LSZ) reduction turns the residues of the external poles into the scattering matrix, and the Källén–Lehmann representation places that pole inside the complete spectrum seen by the chosen operator. These results are what allow statements about the local operator algebra to be translated into statements about observable particles and scattering amplitudes.

They also explain an important independence. A particle need not be tied to one preferred field. Different elementary or composite operators may create the same asymptotic state, with different off-shell Green functions and different one-particle overlaps. After the one-particle normalization has been fixed, however, the physical mass and the S -matrix are unchanged. In algebraic language, within a fixed physical representation of the local algebra, these observables do not depend on which set of local generators is used to describe it. We shall first see this concretely with local fields and only then return to the algebraic statement.

Ultraviolet divergences have not entered. They will matter later, when perturbation theory is

used to calculate the quantities just defined and a regulator is introduced. But the definitions themselves come from the spectrum and from scattering. They make sense at finite cutoff as well as in a continuum theory.

Part I develops this connection from the ground up. We start with a stable scalar particle and a local operator that creates it. We then construct in- and out-states, derive the exact pole factorization, recover LSZ by sequential pole reduction, and finally use the Källén–Lehmann representation to see how the one-particle pole sits inside the full spectrum.

1. Local fields, physical particles and the scattering matrix

We use the mostly-minus metric $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$, write $p^2 = p_0^2 - \mathbf{p}^2$, and define $\square = \partial_\mu \partial^\mu$.

1.1 A stable scalar particle and a local field

Let $|p\rangle$ be an exact one-particle state of mass M , normalized covariantly,

$$\langle p'|p\rangle = 2E_{\mathbf{p}}(2\pi)^3\delta^{(3)}(\mathbf{p} - \mathbf{p}'), \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + M^2}. \quad (1)$$

The mass M here is the exact one-particle mass: it labels an irreducible one-particle representation of the Poincaré group. In the next sections we shall prove that this same M is the position of the isolated one-particle pole of exact correlation functions. We reserve the symbol m for a Lagrangian mass parameter when perturbation theory is introduced later.

Now take any local scalar operator $\Phi(x)$ with the quantum numbers of this particle and suppose

$$\langle \Omega | \Phi(0) | p \rangle \neq 0. \quad (2)$$

We shall say that such an operator *interpolates* the particle, or simply call it an *interpolating operator*: acting on the vacuum, it has a non-zero component along that one-particle state. Because both the vacuum and Φ are Lorentz scalars, Lorentz invariance allows no non-trivial dependence on the on-shell momentum. Thus

$$\langle \Omega | \Phi(0) | p \rangle = C_\Phi, \quad (3)$$

where C_Φ is a number. The operator may be elementary, composite, or something more complicated. Nothing in this statement knows how it was written in a Lagrangian.

From this particular operator we can construct its normalized version,

$$\hat{\Phi}(x) \equiv C_\Phi^{-1} \Phi(x). \quad (4)$$

It satisfies

$$\boxed{\langle \Omega | \hat{\Phi}(0) | p \rangle = 1.} \quad (5)$$

Translation invariance then gives

$$\langle \Omega | \hat{\Phi}(x) | p \rangle = e^{-ip \cdot x}. \quad (6)$$

For the scalar discussion below we denote one such physically normalized interpolator by ϕ_R . Thus ϕ_R is not the name of a unique microscopic field; it denotes a chosen local interpolator whose one-particle overlap has been normalized to unity.

We shall call an interpolating field with this normalization a physically normalized, or renormalized, interpolating field. For now, ‘renormalized’ simply means normalized on the physical one-particle state. The ultraviolet problem will enter later, when we ask how this normalization is calculated in a perturbative representation.

Different local operators can have the same normalized one-particle component without being equal as operators. Their multiparticle matrix elements and short-distance structure may be different. We will return to this distinction when perturbative LSZ is formulated; for the moment all that matters is that a normalized interpolator creates the standard one-particle state.

1.2 The Klein–Gordon inner product and a one-particle wave packet

Scattering states are localized wave packets rather than exact plane waves. We therefore replace the plane wave by a positive-frequency packet whose components all lie on the physical mass shell; separated packets are what allow the asymptotic particle interpretation at early and late times.

Let $f(x)$ be a positive-frequency solution of the free Klein–Gordon equation with the *physical* mass M ,

$$(\square + M^2)f(x) = 0, \quad (7)$$

with

$$f(x) = \int \frac{d^3p}{(2\pi)^3 2E_p} f(\mathbf{p}) e^{-ip \cdot x}. \quad (8)$$

For two complex solutions f and g of the free Klein–Gordon equation define

$$(f, g)_{\text{KG}} = i \int_{\Sigma} d\Sigma^\mu f^*(x) \overleftrightarrow{\partial}_\mu g(x), \quad (9)$$

where

$$f^* \overleftrightarrow{\partial}_\mu g = f^*(\partial_\mu g) - (\partial_\mu f^*)g. \quad (10)$$

The current in (9) is conserved when both entries satisfy the Klein–Gordon equation. Hence the inner product is independent of the Cauchy surface Σ . On a constant-time surface,

$$(f, g)_{\text{KG}} = i \int d^3x f^*(x) \overleftrightarrow{\partial}_0 g(x). \quad (11)$$

For positive-frequency solutions this is positive definite. We shall normalize a one-particle packet

by

$$(f, f)_{\text{KG}} = 1. \quad (12)$$

With the convention (8), this is equivalent to

$$\int \frac{d^3 p}{(2\pi)^3 2E_p} |f(\mathbf{p})|^2 = 1. \quad (13)$$

For a free scalar field, the Klein–Gordon product with a positive-frequency packet simply picks out the corresponding annihilation operator. This suggests defining, for the interacting field,

$$A_f(t) \equiv i \int_{x^0=t} d^3 x f^*(x) \overleftrightarrow{\partial}_0 \phi_R(x). \quad (14)$$

There is one point worth stating carefully. The Klein–Gordon product is conserved when both entries obey the free Klein–Gordon equation. The interacting field ϕ_R does not. Thus $A_f(t)$ is not constant in time. It is precisely its behaviour as $t \rightarrow \pm\infty$ that matters.

The normalization (5) guarantees that the one-particle part of $A_f(t)$ is the same as for a free field. Indeed,

$$\langle \Omega | A_f(t) | p \rangle = i \int d^3 x f^*(x) \overleftrightarrow{\partial}_0 \langle \Omega | \phi_R(x) | p \rangle \quad (15)$$

$$= i \int d^3 x f^*(x) \overleftrightarrow{\partial}_0 e^{-ip \cdot x}. \quad (16)$$

The right-hand side is exactly the Klein–Gordon overlap of the packet with the one-particle plane wave.

The rest of ϕ_R creates multiparticle states as well. For a stable particle with an isolated mass shell, those other contributions separate from the chosen one-particle packet at large times. In the rigorous Haag–Ruelle construction one chooses smooth packets, and for multiparticle states packets with disjoint velocity supports, so that the particles separate as $|t|$ becomes large. Under the usual assumptions of scattering theory one obtains the weak limits

$$\lim_{t \rightarrow -\infty} A_f(t) = a_{\text{in}}(f), \quad \lim_{t \rightarrow +\infty} A_f(t) = a_{\text{out}}(f), \quad (17)$$

and similarly

$$\lim_{t \rightarrow -\infty} A_f^\dagger(t) = a_{\text{in}}^\dagger(f), \quad \lim_{t \rightarrow +\infty} A_f^\dagger(t) = a_{\text{out}}^\dagger(f). \quad (18)$$

The content of these equations is simple. The local field contains many possible excitations. A packet built on an isolated one-particle mass shell filters out the part which travels as that particle, and the normalization of the vacuum-to-one-particle matrix element fixes the normalization of the asymptotic creation and annihilation operators.

1.3 In-states, out-states and the formal definition of the S -matrix

It is useful to state the scattering problem formally before reducing it to correlation functions. Let $\mathcal{H}_0$ denote the asymptotic Fock space built from the stable particle species of the theory. The Møller maps send a free asymptotic state into the corresponding exact state of the interacting theory,

$$\Omega_{\text{in}} : \mathcal{H}_0 \longrightarrow \mathcal{H}, \quad \Omega_{\text{out}} : \mathcal{H}_0 \longrightarrow \mathcal{H}. \quad (19)$$

Formally one may think of them as

$$\Omega_{\text{in}} = \lim_{t \rightarrow -\infty} e^{iHt} e^{-iH_0 t}, \quad \Omega_{\text{out}} = \lim_{t \rightarrow +\infty} e^{iHt} e^{-iH_0 t}, \quad (20)$$

with the usual understanding that in relativistic QFT the existence of scattering states is established more carefully by Haag–Ruelle theory rather than by treating (20) as a literal norm limit on the full Hilbert space.

If $|\alpha\rangle_0$ and $|\beta\rangle_0$ are asymptotic Fock states, define

$$|\alpha, \text{in}\rangle = \Omega_{\text{in}} |\alpha\rangle_0, \quad |\beta, \text{out}\rangle = \Omega_{\text{out}} |\beta\rangle_0. \quad (21)$$

The scattering operator acting on the asymptotic Fock space is

$$S = \Omega_{\text{out}}^\dagger \Omega_{\text{in}}. \quad (22)$$

Thus

$${}_{\text{out}}\langle\beta|\alpha, \text{in}\rangle = {}_0\langle\beta|S|\alpha\rangle_0. \quad (23)$$

For a scalar packet f , the asymptotic creation operators constructed in (18) generate these states:

$$|f_1, \dots, f_n; \text{in}\rangle = a_{\text{in}}^\dagger(f_1) \cdots a_{\text{in}}^\dagger(f_n) |\Omega\rangle, \quad (24)$$

and similarly for the out-state.

The formal definition (22) is useful because it separates two questions. The first is whether asymptotic particle states exist. The second is how to calculate their overlaps from local quantum fields. The normalized interpolating field answers the second question.

1.4 The S -matrix from time-ordered correlation functions

Take an s -particle incoming packet and an r -particle outgoing packet. Their scattering amplitude is

$$\begin{aligned} & \langle g_1, \dots, g_r; \text{out} | f_1, \dots, f_s; \text{in} \rangle \\ &= \langle \Omega | a_{\text{out}}(g_r) \cdots a_{\text{out}}(g_1) a_{\text{in}}^\dagger(f_1) \cdots a_{\text{in}}^\dagger(f_s) | \Omega \rangle. \end{aligned} \quad (25)$$

Using (17) and (18), this can be written entirely in terms of the interacting field:

$$\begin{aligned} & \langle g_1, \dots, g_r; \text{out} | f_1, \dots, f_s; \text{in} \rangle \\ &= \lim_{\substack{t'_1, \dots, t'_r \rightarrow +\infty \\ t_1, \dots, t_s \rightarrow -\infty}} \langle \Omega | A_{g_r}(t'_r) \cdots A_{g_1}(t'_1) A_{f_1}^\dagger(t_1) \cdots A_{f_s}^\dagger(t_s) | \Omega \rangle. \end{aligned} \quad (26)$$

The outgoing times can be taken to the far future with a small ordering among them, and the incoming times to the far past in the same way. Their relative ordering is then precisely the ordering supplied by a time-ordered product.

Substituting (14) gives the surface formula

$$\begin{aligned} & \langle g_1, \dots, g_r; \text{out} | f_1, \dots, f_s; \text{in} \rangle \\ &= \lim_{\substack{t'_a \rightarrow +\infty \\ t_b \rightarrow -\infty}} \left[\prod_{a=1}^r i \int_{x_a^0=t'_a} d^3 x_a g_a^*(x_a) \overleftrightarrow{\partial}_{x_a^0} \right] \\ & \quad \times \left[\prod_{b=1}^s (-i) \int_{y_b^0=t_b} d^3 y_b f_b(y_b) \overleftrightarrow{\partial}_{y_b^0} \right] \\ & \quad \times \langle \Omega | T \{ \phi_R(x_1) \cdots \phi_R(x_r) \phi_R(y_1) \cdots \phi_R(y_s) \} | \Omega \rangle, \end{aligned} \quad (27)$$

where each bidirectional derivative acts between the wave packet and its corresponding field insertion. Here and below the *connected* correlation function means the cumulant: the part which cannot be written as a product of lower-point vacuum correlation functions. Disconnected pieces contain the identity contribution and spectator propagation; the connected part contains the connected scattering amplitude.

Equation (27) is the first main result. The scattering matrix is contained in time-ordered correlation functions of a local field whose one-particle matrix element has been normalized to unity.

It is worth being precise about what has, and has not, happened. We have not used perturbation theory. We have not drawn a diagram. We have not introduced a bare field or a wavefunction counterterm. We have not yet made the diagrammatic statement that an external propagator is to be removed. The only input was the existence of an isolated stable particle, a local operator which couples to it, and the asymptotic construction of particle states.

One may continue from (27) and convert the large-time surface terms into spacetime integrals involving $(\square + M^2)\phi_R$. That is the familiar non-perturbative reduction formula. We shall return to that form after the analytic structure of the correlator has been understood. For the moment, the surface form keeps the physics visible: a detector sees the part of a local field which survives as a freely propagating packet at large time.

1.5 Elementary and composite interpolators

The same construction applies to a composite local operator. Suppose $\mathcal{O}(x)$ is a scalar composite operator and

$$\langle \Omega | \mathcal{O}(0) | p \rangle = C_{\mathcal{O}} \neq 0. \quad (28)$$

Then

$$\hat{\mathcal{O}} = C_{\mathcal{O}}^{-1} \mathcal{O} \quad (29)$$

has unit one-particle overlap and may be used in the same wave-packet construction. Correlation functions of $\hat{\mathcal{O}}$ therefore give the same on-shell scattering matrix as correlation functions of any other correctly normalized interpolator for that particle.

This does not mean that the fields are equal. We shall use square brackets, as in $[\mathcal{O}]_R$, for a UV-renormalized composite operator; such an operator may involve mixing with other local operators. For example, in a $\mathbb{Z}_2$ -invariant scalar theory both ϕ and a properly defined composite operator such as $[\phi^3]_R$ can have a one-particle component with the same quantum numbers. Their short-distance behaviour, their multiparticle matrix elements and, as we shall see in the Källén–Lehmann representation, their continuum spectral densities can all differ. What scattering theory identifies is their common projection onto the same asymptotic one-particle state.

This observation will become important later. Textbooks often write relations of the form $\Phi = Z^{1/2} \Phi_R$ without saying what the equality means. Such an equation is harmless when Φ_R is *defined* to be a rescaled version of that same regulated field. It is imprecise when it is read as an exact relation between an arbitrary perturbative Heisenberg operator and an independently chosen physically normalized interpolator, with the same Z then identified with the LSZ residue. For scattering, the representation-independent content is the form of the one-particle matrix element,

$$\langle \Omega | \Phi_A(0) | p, \sigma \rangle = C_{\Phi} u_A(p, \sigma), \quad (30)$$

with one coefficient C_{Φ} for a non-degenerate particle species, not, in general, an equality between the full interacting operators. Later we shall identify $|C_{\Phi}|^2$ with the pole weight of this chosen operator. The distinction is especially unavoidable for composite operators.

1.6 Bosons with spin

For a particle with spin, the scalar number 1 in (5) is replaced by the standard free one-particle wavefunction. Let $\Phi_{R,A}$ carry Lorentz indices collectively denoted by A . We normalize it by

$$\boxed{\langle \Omega | \Phi_{R,A}(0) | p, \sigma \rangle = u_A(p, \sigma).} \quad (31)$$

The wave packet is built from the same $u_A(p, \sigma)$. At the level of the one-particle Hilbert space its inner product is

$$(f, g)_1 = \sum_{\sigma} \int \frac{d^3 p}{(2\pi)^3 2E_p} f_{\sigma}^*(\mathbf{p}) g_{\sigma}(\mathbf{p}), \quad (32)$$

with the obvious modifications for massless helicity representations. The conserved spacetime symplectic form of the corresponding free field reproduces this inner product and is what is used to project the interacting interpolator onto an asymptotic packet. The rest of the argument is unchanged.

For a massive vector particle,

$$\langle \Omega | V_\mu^R(0) | p, \lambda \rangle = \epsilon_\mu(p, \lambda), \quad p^\mu \epsilon_\mu(p, \lambda) = 0, \quad (33)$$

with $\lambda = 1, 2, 3$. A packet may be written

$$f_\mu(x) = \sum_{\lambda=1}^3 \int \frac{d^3p}{(2\pi)^3 2E_p} f_\lambda(\mathbf{p}) \epsilon_\mu(p, \lambda) e^{-ip \cdot x}. \quad (34)$$

For a Proca solution h_μ define $F_h^{\mu\nu} = \partial^\mu h^\nu - \partial^\nu h^\mu$. On the transverse Proca solution space the conserved product may be written

$$(f, g)_P = i \int_\Sigma d\Sigma_\mu \left[f_\nu^* F_g^{\mu\nu} - F_f^{*\mu\nu} g_\nu \right]. \quad (35)$$

It reproduces the one-particle inner product in Eq. (32) and projects V_μ^R onto the asymptotic annihilation operator for the packet. Thus nothing essential depends on spin. What changes is only the covariant wavefunction carried by the external particle.

1.7 Fermions

For a stable Dirac particle, Lorentz covariance fixes the vacuum-to-one-particle matrix element to be proportional to the free Dirac spinor. We use $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, $\not{p} \equiv \gamma^\mu p_\mu$, and $\bar{u} \equiv u^\dagger \gamma^0$ (similarly for v). The state $|\bar{p}, s\rangle$ denotes the corresponding one-antiparticle state. The physically normalized fermion field is defined by

$$\boxed{\langle \Omega | \psi_R(0) | p, s \rangle = u(p, s),} \quad (36)$$

and for the antiparticle by

$$\boxed{\langle \bar{p}, s | \psi_R(0) | \Omega \rangle = v(p, s).} \quad (37)$$

Equivalently,

$$\langle p, s | \bar{\psi}_R(0) | \Omega \rangle = \bar{u}(p, s), \quad \langle \Omega | \bar{\psi}_R(0) | \bar{p}, s \rangle = \bar{v}(p, s). \quad (38)$$

We use

$$(\not{p} - M)u(p, s) = 0, \quad (\not{p} + M)v(p, s) = 0, \quad (39)$$

and the usual spin sums

$$\sum_s u(p, s) \bar{u}(p, s) = \not{p} + M, \quad \sum_s v(p, s) \bar{v}(p, s) = \not{p} - M. \quad (40)$$

The analogue of the Klein–Gordon product is even simpler. If f and g are solutions of the free

Dirac equation, then

$$(f, g)_D = \int_{\Sigma} d\Sigma_{\mu} \bar{f} \gamma^{\mu} g \quad (41)$$

is independent of the Cauchy surface. On a constant-time slice,

$$(f, g)_D = \int d^3x f^{\dagger}(x) g(x). \quad (42)$$

Choose a positive-energy packet

$$f(x) = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_p} f_s(\mathbf{p}) u(p, s) e^{-ip \cdot x}, \quad (i\not{D} - M)f = 0. \quad (43)$$

Define

$$A_f(t) = \int_{x^0=t} d^3x f^{\dagger}(x) \psi_R(x). \quad (44)$$

Because of (36), its one-particle component has the standard free normalization, and scattering theory gives

$$A_f(t) \longrightarrow \begin{cases} a_{\text{in}}(f), & t \rightarrow -\infty, \\ a_{\text{out}}(f), & t \rightarrow +\infty. \end{cases} \quad (45)$$

For a negative-frequency packet

$$g(x) = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_p} g_s(\mathbf{p}) v(p, s) e^{+ip \cdot x}, \quad (46)$$

the same Dirac product projects ψ_R onto the corresponding antiparticle creation operator; its Hermitian conjugate gives antiparticle annihilation. Thus $u, \bar{u}, v, \bar{v}$ enter the scattering formalism before any Feynman rule is invoked. They are simply the covariant wavefunctions which describe the one-particle parts of the interpolating fields.

The fermionic S -matrix is then obtained from large-time limits of time-ordered correlators of ψ_R and $\bar{\psi}_R$, exactly as in the scalar case, with the familiar minus signs coming from the reordering of fermionic operators. A composite spinor operator is equally admissible. If, for example,

$$\langle \Omega | [\phi\psi](0) | p, s \rangle = C u(p, s), \quad (47)$$

then $C^{-1}[\phi\psi]$ is a correctly normalized interpolator for that same fermion, even though its off-shell correlation functions are very different from those of ψ_R .

1.8 The $U(1)$ gauge field

A gauge field requires one qualification: the local potential contains unphysical gauge components, whereas the photon has only two physical helicities. In a fixed gauge we normalize the physical one-photon matrix element by

$$\boxed{\langle \Omega | A_{\mu}^R(0) | \gamma(p, \lambda) \rangle = \epsilon_{\mu}(p, \lambda), \quad \lambda = 1, 2,} \quad (48)$$

with

$$p^2 = 0, \quad p^\mu \epsilon_\mu(p, \lambda) = 0. \quad (49)$$

A physical photon packet is therefore

$$f_\mu(x) = \sum_{\lambda=1}^2 \int \frac{d^3p}{(2\pi)^3 2|\mathbf{p}|} f_\lambda(\mathbf{p}) \epsilon_\mu(p, \lambda) e^{-ip \cdot x}. \quad (50)$$

For any vector field h_μ write $F_h^{\mu\nu} = \partial^\mu h^\nu - \partial^\nu h^\mu$. For two free Maxwell solutions one may use the conserved symplectic product

$$(f, A)_M = i \int_\Sigma d\Sigma_\mu \left[f_\nu^* F_A^{\mu\nu} - F_f^{*\mu\nu} A_\nu \right], \quad (51)$$

with suitable fall-off conditions. On the physical transverse solution space this projects A_μ^R onto the photon annihilation operator. The asymptotic construction therefore proceeds as before, with external photon amplitudes obtained after projection onto $\epsilon_\mu(p, \lambda)$.

The freedom in the choice of interpolator is particularly transparent here. The gauge-invariant field strength also creates the photon:

$$\langle \Omega | F_{\mu\nu}(0) | \gamma(p, \lambda) \rangle = -i [p_\mu \epsilon_\nu(p, \lambda) - p_\nu \epsilon_\mu(p, \lambda)], \quad (52)$$

up to the Fourier convention. Thus A_μ , after projection onto the physical sector, and $F_{\mu\nu}$ are different local ways of interpolating the same asymptotic photon.

There is an infrared subtlety which should not be confused with the present discussion. In full QED an electrically charged state is accompanied by arbitrarily soft photons, and the naive isolated one-particle pole for a charged field is replaced by the infraparticle structure. We shall postpone this issue. It is a qualification of the asymptotic spectrum, not a change in the logic of how an isolated stable particle is extracted when such a mass shell exists.

1.9 What has been established

The argument so far is non-perturbative. A stable particle is an exact state. A local operator is useful for scattering if it has a non-zero matrix element between the vacuum and that state. Lorentz covariance fixes the form of this matrix element, and we normalize its coefficient so that the local operator has the same one-particle normalization as the corresponding free field. A wave packet, together with the conserved inner product of the free one-particle equation, then isolates the asymptotic particle at early or late times. These asymptotic operators build the in- and out-states, and their overlap is the S -matrix.

For a scalar the normalization is

$$\langle \Omega | \phi_R(0) | p \rangle = 1; \quad (53)$$

for a Dirac particle it is

$$\langle \Omega | \psi_R(0) | p, s \rangle = u(p, s), \quad (54)$$

and for a physical photon

$$\langle \Omega | A_\mu^R(0) | p, \lambda \rangle = \epsilon_\mu(p, \lambda). \quad (55)$$

The corresponding time-ordered correlation functions contain the full scattering matrix.

Nothing in this section has required a weak-coupling expansion. We have not introduced a bare mass, a self-energy, a counterterm, or a diagrammatic external-line amputation. Nor have we assumed that the interpolating operator is elementary. Those ingredients belong to a later, perturbative discussion.

The next question is analytic rather than asymptotic. If a local operator can create a stable particle, how does that particle announce itself inside an exact momentum-space correlation function? The answer is a pole. Establishing that statement, and learning how its residue factorizes, is the subject of polology. The Källén–Lehmann representation will then make two notions precise before perturbation theory begins: the physical mass as the position of the isolated pole, and the one-particle weight of a chosen interpolating operator as its residue.

2. Exact polology: how a particle appears in a correlation function

The preceding section began with particles at large times. We now ask the same question in momentum space. Suppose an exact correlation function is regarded as a function of the momentum carried between two groups of operator insertions. What happens when that momentum can be carried by a real, stable particle of the theory?

The answer is an isolated pole. Its position is the mass of the particle, and its residue factorizes into two exact matrix elements: one describing how the particle is produced by the operators on one side, the other how it is absorbed by the operators on the other. This statement is independent of perturbation theory. There are no Feynman graphs in the argument, and no distinction between elementary and composite particles. What matters is the spectrum of the exact Hamiltonian.

We shall prove the result first for a massive scalar particle. The proof is worth doing carefully. Much of the later diagrammatic language of quantum field theory is a compact way of representing this exact statement.

2.1 The channel variable

Consider the connected time-ordered correlation function

$$\mathcal{G}(p_1, \dots, p_n) = \int \prod_{i=1}^n d^4x_i \exp\left(i \sum_{i=1}^n p_i \cdot x_i\right) \langle \Omega | T\{\mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n)\} | \Omega \rangle_c. \quad (56)$$

Translation invariance gives

$$\mathcal{G}(p_1, \dots, p_n) = (2\pi)^4 \delta^{(4)}\left(\sum_{i=1}^n p_i\right) G(p_1, \dots, p_n), \quad (57)$$

and we shall discuss the reduced function G .

Choose a non-trivial partition of the operators into two groups,

$$\mathcal{A} = \{\mathcal{O}_1, \dots, \mathcal{O}_r\}, \quad \mathcal{B} = \{\mathcal{O}_{r+1}, \dots, \mathcal{O}_n\}, \quad (58)$$

and define the momentum carried from one group to the other by

$$q = \sum_{i=1}^r p_i = - \sum_{j=r+1}^n p_j. \quad (59)$$

We shall keep all other independent momentum variables fixed and study G as a function of q .

To expose the variable conjugate to q , write the coordinates in the two groups as

$$x_i = X + \xi_i \quad (i \leq r), \quad x_j = Y + \eta_j \quad (j > r), \quad (60)$$

where one relative coordinate in each group may be set to zero. The overall translation $(X+Y)/2$ produces the delta function in Eq. (57); the remaining collective coordinate is

$$R = X - Y. \quad (61)$$

After the overall delta function has been removed, the dependence on the channel momentum has the form

$$G = \int [d\xi] [d\eta] e^{i\Phi(\xi, \eta)} \int d^4 R e^{iq \cdot R} \mathcal{G}_R(R; \xi, \eta), \quad (62)$$

where Φ contains only the momenta internal to the two groups. The precise choice of relative coordinates is immaterial. The important point is that q is Fourier conjugate to the separation of the two clusters.

2.2 The two asymptotic time-ordering regions

For fixed ξ_i and η_j , there is a finite time after which every point in the first group lies later than every point in the second. Define

$$T_+(\xi, \eta) = \max_{j>r} \eta_j^0 - \min_{i \leq r} \xi_i^0, \quad (63)$$

$$T_-(\xi, \eta) = \min_{j>r} \eta_j^0 - \max_{i \leq r} \xi_i^0. \quad (64)$$

Then

$$R^0 > T_+ \quad (65)$$

places all operators in $\mathcal{A}$ later than all operators in $\mathcal{B}$, whereas

$$R^0 < T_- \quad (66)$$

places all operators in $\mathcal{B}$ later than all operators in $\mathcal{A}$.

The interval

$$T_- < R^0 < T_+ \quad (67)$$

is finite for fixed relative coordinates. In this interval the time orderings of the two groups can interlace. We study a neighbourhood of the isolated surface $q^2 = M^2$ while keeping the remaining independent momenta away from other singular surfaces; equivalently, one may first smear them with smooth test functions. With this understood, the Fourier transform of the finite R^0 interval is analytic in q^0 near the pole. Subsequent integrations over the internal relative coordinates do not manufacture the isolated q -channel pole so long as no other singular surface is allowed to collide with it. The one-particle pole must therefore come from one of the two infinite tails $R^0 > T_+$ or $R^0 < T_-$. This observation is the small technical point that makes the argument precise: the isolated pole is an asymptotic-time effect.

Consider the first tail. There the time-ordered product factorizes into time ordering within the two groups,

$$\begin{aligned} \mathcal{G}_R(R; \xi, \eta) &= \langle \Omega | T \{ \mathcal{O}_1(X + \xi_1) \cdots \mathcal{O}_r(X + \xi_r) \} \\ &\quad \times T \{ \mathcal{O}_{r+1}(Y + \eta_{r+1}) \cdots \mathcal{O}_n(Y + \eta_n) \} | \Omega \rangle_c. \end{aligned} \quad (68)$$

Insert a complete set of exact energy-momentum eigenstates between the two factors.

Suppose the channel contains an isolated stable scalar particle of mass M . The corresponding one-particle part of the identity is

$$\mathbf{1}_{1p} = \int \frac{d^3k}{(2\pi)^3 2E_k} |k\rangle \langle k|, \quad E_k = \sqrt{\mathbf{k}^2 + M^2}. \quad (69)$$

Its contribution to Eq. (68) has the R -dependence

$$\int \frac{d^3k}{(2\pi)^3 2E_k} e^{-ik \cdot R} A(k; \xi) B(k; \eta), \quad (70)$$

where

$$A(k; \xi) \equiv \langle \Omega | T \{ \mathcal{O}_1(\xi_1) \cdots \mathcal{O}_r(\xi_r) \} | k \rangle, \quad (71)$$

$$B(k; \eta) \equiv \langle k | T \{ \mathcal{O}_{r+1}(\eta_{r+1}) \cdots \mathcal{O}_n(\eta_n) \} | \Omega \rangle. \quad (72)$$

Translations have been used to place the collective coordinates at the origin inside these matrix elements.

The spatial integral over $\mathbf{R}$ is now elementary:

$$\int d^3R e^{-i\mathbf{q}\cdot\mathbf{R}} e^{+i\mathbf{k}\cdot\mathbf{R}} = (2\pi)^3 \delta^{(3)}(\mathbf{q} - \mathbf{k}). \quad (73)$$

Thus the intermediate particle is forced to carry the spatial momentum $\mathbf{q}$. The remaining time integral over the positive tail is

$$\int_{T_+}^{\infty} dR^0 e^{i(q^0 - E_{\mathbf{q}})R^0} = \frac{i e^{i(q^0 - E_{\mathbf{q}})T_+}}{q^0 - E_{\mathbf{q}} + i0}. \quad (74)$$

The finite lower limit appears only in the phase. At the pole,

$$q^0 \longrightarrow E_{\mathbf{q}}, \quad (75)$$

that phase tends to unity. Therefore the residue is independent of the arbitrary finite time needed to separate the two groups.

Near the positive-energy mass shell,

$$q^2 - M^2 = (q^0 - E_{\mathbf{q}})(q^0 + E_{\mathbf{q}}), \quad (76)$$

so that

$$\frac{1}{2E_{\mathbf{q}}} \frac{i}{q^0 - E_{\mathbf{q}} + i0} \simeq \frac{i}{q^2 - M^2 + i0} \quad (77)$$

when the singular part is isolated. After the remaining integrations over ξ and η have been performed, we arrive at the pole formula

$$G(p_1, \dots, p_n) = \frac{i}{q^2 - M^2 + i0} \mathcal{A}(q; \{p_i\}_{i \leq r}) \mathcal{B}(q; \{p_j\}_{j > r}) + G_{\text{reg}}, \quad (78)$$

where G_{reg} is regular at this isolated pole. The two factors in the residue are the Fourier transforms of the exact matrix elements in Eq. (72), evaluated with the intermediate momentum on shell.

This last qualification matters. Polology determines

$$\text{Res}_{q^2=M^2} G, \quad (79)$$

not a preferred off-shell continuation of $\mathcal{A}$ or $\mathcal{B}$. The factors in the residue are intrinsically on-shell quantities. Writing them as functions of an off-shell four-vector is only notation unless some further definition is supplied.

The negative-time tail $R^0 < T_-$ gives the corresponding antiparticle contribution. For a neutral scalar the two descriptions are related in the usual way and give the same invariant mass-shell singularity. For charged particles one keeps track of the charge flowing through the channel; the positive- and negative-energy poles then refer to particle and antiparticle propagation respectively.

Equation (78) is the central result of polology. It has used only translation invariance, complete-

ness of the exact states, and the existence of an isolated one-particle mass shell.

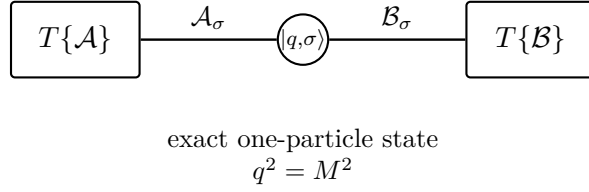


Figure 1: The factorization behind exact polology. This is a Hilbert-space decomposition obtained by inserting exact states between two groups of local operators; it is *not* a perturbative Feynman diagram. An isolated one-particle state produces the simple pole, and its residue is the product of the two exact on-shell matrix elements.

2.3 What the pole formula is saying

There are two distinct statements in Eq. (78).

First, the pole occurs at the mass of an exact state,

$$q^2 = M^2. \quad (80)$$

The mass has not been read from a Lagrangian and has not been corrected by a Feynman graph. It is the mass that labels the one-particle representation of the exact theory.

Second, the residue factorizes. A state which propagates for arbitrarily long time between the two groups carries no memory of the detailed mechanism by which it was created. The left group contributes only its matrix element for producing the particle; the right group contributes only its matrix element for absorbing it. The universality of a particle is encoded in the factorization of the pole.

A continuum behaves differently. For fixed total spatial momentum, a two-particle state has a continuous range of energies because momentum may be shared between the constituents. The analogue of Eq. (70) then contains an integral over a continuum of invariant masses. Instead of an isolated pole one obtains a threshold singularity and, after analytic continuation, a branch cut. We shall see this in its cleanest form in the Källén–Lehmann representation.

This also explains why bound states cause no conceptual difficulty. If the exact spectrum contains a stable bound state, it produces a pole whether or not a field with its name occurs in the microscopic Lagrangian. Conversely, an unstable resonance is not an exact asymptotic one-particle state. It does not give a delta-function mass shell on the physical sheet; its pole appears only after analytic continuation to another sheet.

2.4 Bosons with spin

The proof changes very little when the intermediate particle carries spin. The one-particle resolution of the identity becomes

$$\mathbf{1}_p = \sum_{\sigma} \int \frac{d^3k}{(2\pi)^3 2E_k} |k, \sigma\rangle \langle k, \sigma|. \quad (81)$$

Repeating the previous steps gives

$$G = \frac{i}{q^2 - M^2 + i0} \sum_{\sigma} \mathcal{A}_{\sigma}(q) \mathcal{B}_{\sigma}(q) + G_{\text{reg}}. \quad (82)$$

The sum is over the physical polarizations of the exact one-particle state.

If a local operator $\Phi_{R,A}$ is normalized by

$$\langle \Omega | \Phi_{R,A}(0) | p, \sigma \rangle = u_A(p, \sigma), \quad (83)$$

then the singular part of its two-point function is

$$G_{AB}(p) \simeq \frac{i}{p^2 - M^2 + i0} \Pi_{AB}(p), \quad \Pi_{AB}(p) = \sum_{\sigma} u_A(p, \sigma) u_B^*(p, \sigma). \quad (84)$$

Thus the numerator of the one-particle pole is simply the completeness relation on the physical polarization space.

For a massive vector field,

$$\langle \Omega | V_{\mu}^R(0) | p, \lambda \rangle = \epsilon_{\mu}(p, \lambda), \quad (85)$$

and

$$\sum_{\lambda=1}^3 \epsilon_{\mu}(p, \lambda) \epsilon_{\nu}^*(p, \lambda) = -\eta_{\mu\nu} + \frac{p_{\mu} p_{\nu}}{M^2} \quad (86)$$

for the mostly-minus metric. The familiar tensor numerator therefore has a direct spectral meaning before any propagator has been introduced perturbatively.

2.5 Fermionic polology

For a stable Dirac particle the normalized interpolating field satisfies

$$\langle \Omega | \psi_R(0) | p, s \rangle = u(p, s), \quad \langle p, s | \bar{\psi}_R(0) | \Omega \rangle = \bar{u}(p, s), \quad (87)$$

and for the antiparticle

$$\langle \bar{p}, s | \psi_R(0) | \Omega \rangle = v(p, s), \quad \langle \Omega | \bar{\psi}_R(0) | \bar{p}, s \rangle = \bar{v}(p, s). \quad (88)$$

The two-point function shows the mechanism most transparently. Define

$$S_{R,\alpha\beta}(x) \equiv \langle \Omega | T \{ \psi_{R,\alpha}(x) \bar{\psi}_{R,\beta}(0) \} | \Omega \rangle. \quad (89)$$

For $x^0 > 0$, insertion of exact one-fermion states gives

$$S_{R,\alpha\beta}(x)|_{1p} = \sum_s \int \frac{d^3k}{(2\pi)^3 2E_k} u_\alpha(k, s) \bar{u}_\beta(k, s) e^{-ik \cdot x}. \quad (90)$$

For $x^0 < 0$, the antiparticle contribution is

$$S_{R,\alpha\beta}(x)|_{1\bar{p}} = - \sum_s \int \frac{d^3k}{(2\pi)^3 2E_k} v_\alpha(k, s) \bar{v}_\beta(k, s) e^{+ik \cdot x}, \quad (91)$$

where the minus sign is the one required by fermionic time ordering. Using

$$\sum_s u(k, s) \bar{u}(k, s) = \not{k} + M, \quad \sum_s v(k, s) \bar{v}(k, s) = \not{k} - M, \quad (92)$$

the singular part of the Fourier transform is

$$S_R(p) \simeq \frac{i(\not{p} + M)}{p^2 - M^2 + i0}. \quad (93)$$

This is the clean pole statement. The numerator is not a perturbative embellishment: it is the spin sum over the exact one-particle states.

The general channel formula is equally simple. If a fermion of mass M and momentum q can propagate between the two groups, then

$$G = \frac{i}{q^2 - M^2 + i0} \sum_s \mathcal{A}_s(q) \mathcal{B}_s(q) + G_{\text{reg}}, \quad (94)$$

with the sign fixed by the chosen ordering convention for the fermionic operators. When the spin dependence on the two sides is represented with open Dirac indices, the same result may be written

$$G \simeq \mathcal{A}^\alpha(q) \frac{i(\not{q} + M)_{\alpha\beta}}{q^2 - M^2 + i0} \mathcal{B}^\beta(q). \quad (95)$$

For an arbitrary spinor operator Ψ ,

$$\langle \Omega | \Psi(0) | p, s \rangle = C_\Psi u(p, s), \quad (96)$$

and each occurrence of that operator in a one-particle residue contributes the corresponding factor C_Ψ . Again, this is a statement about the one-particle projection of the operator, not an equality between full interacting fields.

2.6 The massless $U(1)$ field

A massless gauge field deserves one qualification before the formula is written. The proof above used an isolated massive mass hyperboloid. In a massless theory the continuum can reach $p^2 = 0$, and the Haag–Ruelle mass-gap argument does not apply without modification. Gauge fixing also enlarges the state space by unphysical polarizations. These infrared and gauge-theory issues should not be hidden.

Nevertheless, when a physical asymptotic photon state exists, its one-particle contribution is obtained by projecting onto the two transverse helicities. With

$$\langle \Omega | A_\mu^R(0) | \gamma(p, \lambda) \rangle = \epsilon_\mu(p, \lambda), \quad \lambda = 1, 2, \quad (97)$$

we have, in the physical polarization sector,

$$G_\mu(p; \dots) \simeq \frac{i}{p^2 + i0} \sum_{\lambda=1}^2 \epsilon_\mu(p, \lambda) \mathcal{F}_\lambda(p; \dots), \quad (98)$$

and a physical polarization $\epsilon^{*\mu}(p, \lambda)$ extracts the corresponding residue. For gauge-invariant operators the same physical pole may be exhibited without referring to an unphysical polarization space. For example,

$$\langle \Omega | F_{\mu\nu}(0) | \gamma(p, \lambda) \rangle = -i(p_\mu \epsilon_\nu - p_\nu \epsilon_\mu). \quad (99)$$

We shall return to the special infrared structure of QED when gauge-theory renormalization is discussed.

2.7 Pion exchange: a pole that can be measured without reaching it

Polology becomes useful when the pole lies outside the kinematic region of the process one is studying. The classic example is pion exchange.

Let $\pi_R^a(x)$ be a physically normalized interpolating operator for the pion, with isospin index a , and consider its matrix element between one-nucleon states,

$$\Gamma^a(p', p) = \langle N(p') | \pi_R^a(0) | N(p) \rangle, \quad q = p - p', \quad t = q^2. \quad (100)$$

Lorentz invariance, parity and isospin imply that its one-pion pole may be parametrized by defining the physical pion–nucleon coupling through the residue. Here τ^a are the Pauli matrices acting on the nucleon isospin doublet and $\gamma_5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$:

$$\lim_{t \rightarrow M_\pi^2} (t - M_\pi^2) \Gamma^a(p', p) \equiv g_{\pi NN} \bar{u}(p') i\gamma_5 \tau^a u(p), \quad (101)$$

up to the harmless overall phase convention chosen for the pion interpolating field. Thus, near the pion pole,

$$\Gamma^a(p', p) \simeq \frac{g_{\pi NN}}{t - M_\pi^2 + i0} \bar{u}(p') i\gamma_5 \tau^a u(p) + \text{regular}. \quad (102)$$

For elastic nucleon scattering the physical momentum transfer is spacelike, $t \leq 0$, so the pole at $t = M_\pi^2$ is not itself reached. It is a singularity of the analytically continued matrix element. Yet its position and residue control observable low-momentum physics.

The same pole appears in nucleon–nucleon scattering. Near the one-pion singularity in the t channel, exact factorization gives

$$\mathcal{M}_{NN} \simeq \frac{g_{\pi NN}^2}{t - M_\pi^2 + i0} [\bar{u}'_1 i\gamma_5 \tau^a u_1] [\bar{u}'_2 i\gamma_5 \tau^a u_2] + \text{less singular terms.} \quad (103)$$

This formula is not a model of the short-distance nuclear force. It is the exact form of the amplitude sufficiently close to the pion pole. In the non-relativistic limit the denominator becomes $-(\mathbf{q}^2 + M_\pi^2)$; Fourier transformation gives the familiar Yukawa range

$$\int \frac{d^3q}{(2\pi)^3} \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{\mathbf{q}^2 + M_\pi^2} = \frac{e^{-M_\pi r}}{4\pi r}, \quad (104)$$

while the numerator supplies the spin–isospin and tensor structure of one-pion exchange. The long-range part of the nuclear force is therefore governed by the nearest singularity allowed in the crossed channel. Short-distance physics changes the regular terms and more distant singularities; it cannot move the pion pole.

2.8 A pion pole in the nucleon axial form factor

A *form factor* is a scalar function of momentum invariants which multiplies an allowed Lorentz structure in a matrix element of a local current. It packages the non-trivial momentum dependence once symmetry has fixed the tensor or spinor structures.

The same reasoning gives a particularly clean form-factor application. The isovector axial current creates a pion from the vacuum,

$$\langle \Omega | A_a^\mu(0) | \pi^b(q) \rangle = i f_\pi q^\mu \delta_a^b. \quad (105)$$

Hence the nucleon matrix element of the axial current must contain a pion pole in its longitudinal part. Polology gives directly

$$\langle N(p') | A_a^\mu(0) | N(p) \rangle \supset \frac{i f_\pi q^\mu}{q^2 - M_\pi^2 + i0} g_{\pi NN} \bar{u}(p') i\gamma_5 \tau_a u(p). \quad (106)$$

Thus the induced pseudoscalar form factor is forced to contain a pion pole. The residue factorizes into two independently meaningful quantities: the amplitude for the axial current to create a pion, f_π , and the amplitude for that pion to couple to the nucleon, $g_{\pi NN}$. This pole is central in low-energy weak processes involving nucleons, including muon capture and neutrino reactions. The usefulness of polology is precisely that the singular part is fixed without knowing the full off-shell dynamics of the nucleon.

2.9 The electromagnetic pion form factor and the boundary of the theorem

For a charged pion the electromagnetic current matrix element is conventionally written

$$\langle \pi^+(p') | J_\mu^{\text{em}}(0) | \pi^+(p) \rangle = (p + p')_\mu F_\pi(t), \quad t = (p' - p)^2, \quad (107)$$

with $F_\pi(0) = 1$ when the current is normalized to unit electric charge.

Viewed in the crossed channel, the current carries the quantum numbers of a vector state. If the exact spectrum contained a stable vector particle V of mass M_V with non-zero matrix elements

$$\langle \Omega | J_\mu^{\text{em}}(0) | V(q, \lambda) \rangle \neq 0, \quad \langle V(q, \lambda) | \pi\pi \rangle \neq 0, \quad (108)$$

then the pole formula would require

$$F_\pi(t) \simeq \frac{(\text{current-}V \text{ coupling}) \times (V \rightarrow \pi\pi \text{ coupling})}{t - M_V^2 + i0} + \text{regular}. \quad (109)$$

The real world provides an instructive refinement. The ρ meson is unstable, so it is not an isolated asymptotic state and there is no exact physical-sheet ρ pole of the kind proved above. The pion form factor instead has the two-pion cut beginning at $t = 4M_\pi^2$. Continuing the analytic function through that cut reaches another branch, conventionally called the *second sheet*; the ρ appears there as a resonance pole. The familiar success of vector-meson dominance is therefore not a counterexample to exact polology; it is an approximation in which a nearby resonance dominates the spectral function.

This example is useful because it marks the precise boundary of the theorem. Stable states give physical-sheet poles. Continua give cuts. Resonances reveal themselves only after analytic continuation.

2.10 What has been established

The theorem proved in this section is deliberately more general than a reduction formula. Choose any momentum channel of an exact time-ordered correlation function and separate the operator insertions into two groups. If the exact spectrum contains a stable one-particle state of mass M carrying the quantum numbers and momentum of that channel, then, near its mass shell,

$$G \simeq \frac{i}{q^2 - M^2 + i0} \sum_\sigma \mathcal{A}_\sigma \mathcal{B}_\sigma + \text{terms regular at } q^2 = M^2. \quad (110)$$

The two factors in the residue are exact on-shell matrix elements. The pole position is fixed by the physical mass; the residue records how the chosen operators couple to that state. Stable bound states are included automatically. Multiparticle states give threshold singularities and cuts instead of isolated poles, while unstable resonances appear only after analytic continuation away from the physical sheet.

The examples also show why polology is useful beyond the formal study of Green functions. The

pion pole fixes the longest-range part of the nuclear force; the same pole controls the longitudinal part of the nucleon axial form factor; and the analytic structure of the electromagnetic pion form factor distinguishes a stable-particle pole from a resonance embedded in a continuum.

We now return, deliberately, to scattering. Section 1 obtained the S -matrix from asymptotic wave packets. The next section will recover the same non-perturbative reduction formula in a different way, by applying the pole theorem one external particle at a time. The repetition is intentional: the first derivation explains what scattering states are, while the second explains what the reduction formula is doing analytically.

3. Sequential pole reduction: LSZ seen through polology

Section 1 reached the scattering matrix by following wave packets to the remote past and future. We now derive the same result a second time, starting instead from the pole theorem of Section 2. Nothing perturbative is being introduced. The purpose of the repetition is different: it shows that the reduction formula is, analytically, the repeated extraction of isolated one-particle poles.

The scalar case contains the whole idea. We shall therefore do it carefully and then state the changes required for spin.

3.1 The connected scattering amplitude

Consider a stable neutral scalar particle of physical mass M , and let ϕ_R be a physically normalized interpolating field,

$$\langle \Omega | \phi_R(0) | p \rangle = 1. \quad (111)$$

We use covariantly normalized one-particle states as in Eq. (1). For s incoming particles with momenta $p_1, \dots, p_s$ and r outgoing particles with momenta $p'_1, \dots, p'_r$, define the connected invariant amplitude $\mathcal{M}_{fi}$ by

$${}_{\text{out}} \langle p'_1, \dots, p'_r | p_1, \dots, p_s \rangle_{\text{in}, c} = i(2\pi)^4 \delta^{(4)} \left(\sum_{b=1}^s p_b - \sum_{a=1}^r p'_a \right) \mathcal{M}_{fi}. \quad (112)$$

The subscript c means that the identity contribution and all spectator/disconnected pieces have been removed. The plane-wave formula is distributional. The rigorous scattering statement is obtained after smearing each external momentum with a smooth wave packet of the kind used in Section 1; the plane-wave expression is the convenient shorthand obtained after the packet dependence has been stripped away.

Now define the connected momentum-space correlation function

$$\begin{aligned} \mathcal{G}_{r+s}(\{p'\}; \{p\}) &= \int \prod_{a=1}^r d^4 x_a \prod_{b=1}^s d^4 y_b \exp \left(i \sum_{a=1}^r p'_a \cdot x_a - i \sum_{b=1}^s p_b \cdot y_b \right) \\ &\times \langle \Omega | T \{ \phi_R(x_1) \cdots \phi_R(x_r) \phi_R(y_1) \cdots \phi_R(y_s) \} | \Omega \rangle_c. \end{aligned} \quad (113)$$

Until the final limit is taken, the external four-momenta in Eq. (113) are independent off-shell Fourier variables. The physical momenta satisfy

$$p_a'^2 = p_b^2 = M^2, \quad p_a'^0 > 0, \quad p_b^0 > 0. \quad (114)$$

Translation invariance supplies the same overall momentum-conserving delta function as in Eq. (112).

3.2 Removing one outgoing field insertion

Apply the general pole theorem to the channel in which the first group contains only the operator $\phi_R(x_1)$. Because of Eq. (111), the matrix element on that side of the pole is unity. Therefore, as $p_1'^2 \rightarrow M^2$,

$$\mathcal{G}_{r+s} \simeq \frac{i}{p_1'^2 - M^2 + i0} \mathcal{G}_{r+s-1}^{(1)}(p_1'; p_2', \dots, p_r'; p_1, \dots, p_s) + \text{regular}, \quad (115)$$

where the residue is the Fourier transform of the exact matrix element

$$\langle p_1' | T \{ \phi_R(x_2) \cdots \phi_R(x_r) \phi_R(y_1) \cdots \phi_R(y_s) \} | \Omega \rangle_c. \quad (116)$$

The equality in Eq. (115) is a statement about the singular part near the pole. The object on the right is defined only with p_1' on its physical mass shell. No off-shell continuation of the residue is required.

The operation has a simple interpretation. One local field insertion has disappeared; in its place there is an exact one-particle state. This is reduction in its most literal form.

Apply the same argument to x_2 . The second pole converts the second field insertion into a second outgoing particle,

$$\mathcal{G}_{r+s-1}^{(1)} \simeq \frac{i}{p_2'^2 - M^2 + i0} \mathcal{G}_{r+s-2}^{(2)} + \text{regular at } p_2'^2 = M^2. \quad (117)$$

Continuing in this way gives

$$\mathcal{G}_{r+s} \simeq \left[\prod_{a=1}^r \frac{i}{p_a'^2 - M^2 + i0} \right] \mathcal{H}_r(\{p'\}; \{p\}), \quad (118)$$

where $\mathcal{H}_r$ is an exact matrix element with the r outgoing one-particle states already present in the bra.

The order in which isolated external poles are taken is immaterial provided one stays away from exceptional kinematic configurations in which different singular surfaces collide. With wave packets this statement is particularly transparent: each residue projects onto a distinct isolated one-particle mass shell.

3.3 The incoming particles

The incoming insertions are reduced in the same way, now using the part of the pole theorem in which the interpolating field creates a one-particle state from the vacuum in the remote past. Taking the pole in p_1^2 converts $\phi_R(y_1)$ into the exact incoming particle $|p_1\rangle$; then p_2^2 , and so on. After all external poles have been taken, no local interpolating fields remain. What remains is the connected overlap of the exact out- and in-states, namely the quantity defined in Eq. (112).

Consequently the simultaneous singular part of the connected Green function is

$$\begin{aligned} \mathcal{G}_{r+s}(\{p'\}; \{p\}) &\simeq \left[\prod_{a=1}^r \frac{i}{p_a'^2 - M^2 + i0} \right] \left[\prod_{b=1}^s \frac{i}{p_b^2 - M^2 + i0} \right] \\ &\times i(2\pi)^4 \delta^{(4)} \left(\sum_b p_b - \sum_a p_a' \right) \mathcal{M}_{fi} \\ &+ \text{terms missing at least one external pole.} \end{aligned} \quad (119)$$

Equation (119) is the non-perturbative momentum-space LSZ statement for physically normalized scalar interpolating fields.

Equivalently,

$$\begin{aligned} &i(2\pi)^4 \delta^{(4)} \left(\sum_b p_b - \sum_a p_a' \right) \mathcal{M}_{fi} \\ &= \left[\prod_{a=1}^r \lim_{p_a'^2 \rightarrow M^2} \frac{p_a'^2 - M^2}{i} \right] \left[\prod_{b=1}^s \lim_{p_b^2 \rightarrow M^2} \frac{p_b^2 - M^2}{i} \right] \mathcal{G}_{r+s}(\{p'\}; \{p\}). \end{aligned} \quad (120)$$

The limits in Eq. (120) are residue operations. Regular terms vanish because they are multiplied by a factor that goes to zero; a simple one-particle pole survives with its residue.

This is the same scattering information that was obtained in Section 1 from large-time wave packets. There, the particle was isolated by following a packet to temporal infinity. Here, it is isolated by approaching its mass-shell pole. The two procedures are two descriptions of the same projection onto the one-particle subspace.

3.4 The coordinate-space reduction operator

The familiar differential form of LSZ is now almost automatic. With the Fourier convention in Eq. (113),

$$(\square_x + M^2)e^{+ip \cdot x} = -(p^2 - M^2)e^{+ip \cdot x}, \quad (121)$$

and the same identity holds for $e^{-ip \cdot x}$. Integrating by parts in the sense of distributions gives

$$\frac{p^2 - M^2}{i} \int d^4x e^{\pm ip \cdot x} F(x) = i \int d^4x e^{\pm ip \cdot x} (\square_x + M^2) F(x), \quad (122)$$

provided the external wave packets have the usual fall-off so that boundary terms vanish. When F is a time-ordered distribution, the derivatives in Eq. (122) act on the complete distribution,

including its step functions.

Substituting Eq. (122) into Eq. (120) yields

$$\begin{aligned}
& i(2\pi)^4 \delta^{(4)} \left(\sum_b p_b - \sum_a p'_a \right) \mathcal{M}_{fi} \\
&= \left[\prod_{a=1}^r i \int d^4 x_a e^{+ip'_a \cdot x_a} (\square_{x_a} + M^2) \right] \left[\prod_{b=1}^s i \int d^4 y_b e^{-ip_b \cdot y_b} (\square_{y_b} + M^2) \right] \\
&\quad \times \langle \Omega | T \{ \phi_R(x_1) \cdots \phi_R(x_r) \phi_R(y_1) \cdots \phi_R(y_s) \} | \Omega \rangle_c.
\end{aligned} \tag{123}$$

This is the usual coordinate-space LSZ formula, but its meaning is now transparent. The operator $(\square + M^2)$ is not an arbitrary prescription attached to external particles. In momentum space it is precisely the operation that removes the denominator of the exact one-particle pole.

It is useful to keep one distinction clear. Equation (123) is an exact reduction formula written in terms of the physically normalized interpolating field. It contains no statement about Feynman diagrams, no instruction to “cut an external line”, and no perturbative wavefunction factor. Those belong to the perturbative part of the notes.

3.5 Fermions

For a stable Dirac particle, the physically normalized field satisfies

$$\langle \Omega | \psi_R(0) | p, s \rangle = u(p, s), \quad \langle \bar{p}, s | \psi_R(0) | \Omega \rangle = v(p, s). \tag{124}$$

Section 2 showed that an external fermion pole has the form

$$\tilde{G}_\alpha(p; \dots) \simeq \frac{i(\not{p} + M)_{\alpha\beta}}{p^2 - M^2 + i0} \mathcal{F}^\beta(p; \dots). \tag{125}$$

Multiplication by the free inverse Dirac operator removes the pole because

$$(\not{p} - M)(\not{p} + M) = p^2 - M^2. \tag{126}$$

Hence

$$\lim_{p^2 \rightarrow M^2} \frac{1}{i} (\not{p} - M) \tilde{G}(p; \dots) = \mathcal{F}(p; \dots), \tag{127}$$

with the appropriate open spinor index. The physical spin state is then selected by contracting with $u, \bar{u}, v$, or $\bar{v}$, according to whether the external particle is an incoming or outgoing fermion or antifermion.

In coordinate space the inverse free operator is

$$i\not{\partial} - M, \tag{128}$$

with a left-acting version for a conjugate spinor insertion. Thus the scalar factor $(\square + M^2)$ is replaced by the corresponding inverse free Dirac operator, and the external plane waves are

replaced by the free spinor wavefunctions. The origin of the rule is again the exact one-particle pole and the completeness relation

$$\sum_s u(p, s) \bar{u}(p, s) = \not{p} + M, \quad \sum_s v(p, s) \bar{v}(p, s) = \not{p} - M. \quad (129)$$

3.6 Bosons with spin and the photon

For a stable massive particle of arbitrary spin, let the normalized local interpolator obey

$$\langle \Omega | \Phi_{R,A}(0) | p, \sigma \rangle = u_A(p, \sigma). \quad (130)$$

The exact pole carries the physical polarization projector

$$\Pi_{AB}(p) = \sum_{\sigma} u_A(p, \sigma) u_B^*(p, \sigma). \quad (131)$$

The reduction operation is the inverse of the appropriate free wave operator, followed by projection onto the physical polarization $u_A(p, \sigma)$. The scalar and Dirac formulas are simply the two most familiar cases.

For a photon, one works in the physical transverse sector. If

$$\langle \Omega | A_{\mu}^R(0) | \gamma(p, \lambda) \rangle = \epsilon_{\mu}(p, \lambda), \quad \lambda = 1, 2, \quad (132)$$

then the physical pole of a correlation function containing A_{μ}^R is

$$\begin{aligned} \tilde{G}_{\mu}(p; \dots) &\simeq \frac{i}{p^2 + i0} \sum_{\lambda=1}^2 \epsilon_{\mu}(p, \lambda) \mathcal{F}_{\lambda}(p; \dots) \\ &+ \text{terms with no physical transverse one-photon residue.} \end{aligned} \quad (133)$$

Multiplication by p^2 and contraction with $\epsilon^{*\mu}(p, \lambda)$ extracts the physical one-photon residue. The infrared qualifications stated earlier remain in force: the massless case requires more care than the massive Haag–Ruelle setting, and charged states in QED have the infraparticle complication.

3.7 Why this second derivation is worth keeping

The reduction formula has now appeared twice. In Section 1 it followed from the asymptotic meaning of a normalized local field and from the existence of in- and out-states. In this section it followed from exact polology by removing one-particle pole after one-particle pole. The two derivations answer different questions.

The wave-packet derivation explains why a local field can be used to create and detect asymptotic particles. The pole derivation explains why the same information is encoded in a very specific singular part of a momentum-space correlator. This distinction becomes essential when perturbation theory is introduced. The exact reduction theorem will survive unchanged; what will

change is the relation between a convenient perturbative operator and the physically normalized interpolator.

Before perturbation theory begins, however, there is one more exact result to establish. Polology identified one isolated state at a time. The Källén–Lehmann representation describes, in a single formula, the contribution of the entire spectrum to a two-point function.

4. The Källén–Lehmann representation

Polology answers a local question in the complex momentum plane: what singularity is produced by a particular state that can propagate in a chosen channel? The Källén–Lehmann representation answers a broader question for the two-point function: what is the contribution of every state that a chosen local operator can create from the vacuum?

The result is exact. It uses the same ingredients as polology—translation invariance, completeness, Lorentz invariance and positivity of the physical Hilbert space—but packages them into a spectral measure. For renormalization this representation is decisive, because it makes two notions precise before any weak-coupling calculation is attempted: the physical mass is the position of an isolated one-particle pole, and the one-particle pole weight of a chosen interpolating operator is the weight of that state in the operator’s spectral measure. The pole mass is representation-independent; the pole weight is exact but operator-dependent.

4.1 The Wightman function and the spectral measure

Let $\Phi(x)$ be a Hermitian scalar local operator. If its vacuum expectation value is non-zero, subtract it and define

$$\hat{\Phi}(x) = \Phi(x) - \langle \Omega | \Phi(0) | \Omega \rangle. \quad (134)$$

We shall write Φ for the subtracted operator from now on. Define the Wightman two-point function

$$W(x) = \langle \Omega | \Phi(x) \Phi(0) | \Omega \rangle. \quad (135)$$

Insert a complete set of exact energy–momentum eigenstates. The symbol $\sum_N$ below denotes both sums over discrete quantum numbers and the appropriate integrals over continuous ones:

$$W(x) = \sum_N e^{-ip_N \cdot x} |\langle \Omega | \Phi(0) | N \rangle|^2. \quad (136)$$

Its Fourier transform is therefore

$$\widetilde{W}(q) \equiv \int d^4x e^{iq \cdot x} W(x) = (2\pi)^4 \sum_N \delta^{(4)}(q - p_N) |\langle \Omega | \Phi(0) | N \rangle|^2. \quad (137)$$

Every term on the right is non-negative when tested against a non-negative test function. The spectrum condition places p_N in the forward light cone. Lorentz invariance then implies that the distribution can depend only on q^2 once $q^0 > 0$ has been selected.

We therefore *define* the scalar spectral density $\rho_\Phi(\mu^2)$ by

$$\boxed{\widetilde{W}(q) = 2\pi \theta(q^0) \rho_\Phi(q^2)}. \quad (138)$$

With this normalization a free canonically normalized scalar field of mass M_{free} has

$$\rho_{\text{free}}(\mu^2) = \delta(\mu^2 - M_{\text{free}}^2). \quad (139)$$

Equation (137) implies

$$\rho_\Phi(\mu^2) \geq 0 \quad (140)$$

as a distribution. This positivity is a Hilbert-space statement, not a perturbative one.

Introduce the positive-frequency free scalar function of mass μ ,

$$\Delta_+(x; \mu^2) = \int \frac{d^3k}{(2\pi)^3 2E_{\mathbf{k},\mu}} e^{-ik \cdot x}, \quad E_{\mathbf{k},\mu} = \sqrt{\mathbf{k}^2 + \mu^2}. \quad (141)$$

Equivalently,

$$\Delta_+(x; \mu^2) = \int \frac{d^4q}{(2\pi)^3} \theta(q^0) \delta(q^2 - \mu^2) e^{-iq \cdot x}. \quad (142)$$

Fourier inversion of Eq. (138) gives the first form of the Källén–Lehmann representation,

$$\boxed{W(x) = \int_0^\infty d\mu^2 \rho_\Phi(\mu^2) \Delta_+(x; \mu^2)}. \quad (143)$$

The interacting two-point function is thus a positive superposition of free one-particle Wightman functions with all invariant masses present in the exact spectrum.

4.2 The time-ordered function

Define the exact time-ordered two-point function

$$G_F(x) = \langle \Omega | T \{ \Phi(x) \Phi(0) \} | \Omega \rangle. \quad (144)$$

For a Hermitian scalar operator,

$$G_F(x) = \theta(x^0) W(x) + \theta(-x^0) W(-x). \quad (145)$$

Define the free Feynman function of mass μ by

$$\Delta_F(x; \mu^2) = \int \frac{d^4p}{(2\pi)^4} \frac{i e^{-ip \cdot x}}{p^2 - \mu^2 + i0}. \quad (146)$$

Using Eq. (143) in the two time-ordering regions gives the non-local part of the time-ordered function as a superposition of free Feynman functions,

$$G_F(x) = \int_0^\infty d\mu^2 \rho_\Phi(\mu^2) \Delta_F(x; \mu^2) + \text{local contact terms}. \quad (147)$$

For a canonically normalized elementary scalar the contact polynomial can be absent with the standard definition of the time-ordered product. For a general renormalized composite operator, however, coincident-point singularities allow local terms. In momentum space the most general form is therefore

$$\boxed{\tilde{G}_F(p) = P_\Phi(p^2) + \int_0^\infty d\mu^2 \rho_\Phi(\mu^2) \frac{i}{p^2 - \mu^2 + i0}}, \quad (148)$$

where $P_\Phi(p^2)$ is a local polynomial, or equivalently the expression is understood with the number of subtractions required by the ultraviolet growth of the composite correlator. The polynomial has neither an isolated particle pole nor a branch cut. It therefore changes none of the spectral conclusions below: pole positions, pole weights and discontinuities are fixed by ρ_Φ .

No weak-coupling expansion has entered. Equation (148) is an exact spectral representation, modulo these explicitly local subtraction terms, of the two-point function seen by the operator Φ .

4.3 Stable particles, continua and the one-particle pole weight

Suppose the operator has non-zero overlap with a stable scalar particle of mass M ,

$$\langle \Omega | \Phi(0) | p \rangle = C_\Phi. \quad (149)$$

The contribution of this state to Eq. (137) is

$$2\pi\theta(q^0)|C_\Phi|^2\delta(q^2 - M^2). \quad (150)$$

Define

$$\boxed{Z_\Phi^{\text{pole}} \equiv |C_\Phi|^2}. \quad (151)$$

If this is the only isolated stable state in the channel before the continuum begins, then

$$\rho_\Phi(\mu^2) = Z_\Phi^{\text{pole}}\delta(\mu^2 - M^2) + \rho_{\Phi,c}(\mu^2), \quad (152)$$

where $\rho_{\Phi,c}$ has support beginning at the lowest multiparticle invariant mass s_0 to which Φ couples. More generally, if several stable particles with the same quantum numbers exist, the spectral density contains a sum of delta functions, one for each isolated state.

Substitution into Eq. (148) gives

$$\boxed{\tilde{G}_F(p) = P_\Phi(p^2) + \frac{iZ_\Phi^{\text{pole}}}{p^2 - M^2 + i0} + \int_{s_0}^\infty d\mu^2 \rho_{\Phi,c}(\mu^2) \frac{i}{p^2 - \mu^2 + i0}}. \quad (153)$$

This equation makes two definitions precise:

$$M^2 = \text{position of the isolated one-particle pole}, \quad (154)$$

while

$$Z_{\Phi}^{\text{pole}} = \text{residue of that pole divided by } i = |\langle \Omega | \Phi(0) | p \rangle|^2. \quad (155)$$

The mass belongs to the physical state. The residue belongs to the chosen operator and measures how strongly that operator projects onto the state.

For the physically normalized interpolating field of Section 1,

$$\langle \Omega | \phi_R(0) | p \rangle = 1, \quad (156)$$

so its spectral density contains

$$\rho_R(\mu^2) = \delta(\mu^2 - M^2) + \rho_{R,c}(\mu^2), \quad (157)$$

and its two-point function has unit one-particle residue.

A point made earlier is now visible in the spectral representation. Two local operators may interpolate the same particle and be normalized to the same one-particle matrix element, yet have different continuum spectral densities. Equality of their one-particle projections does not make them the same operator.

4.4 The cut and the spectral density

The continuum term in Eq. (153) produces a branch cut along the real p^2 axis beginning at s_0 . To state the relation without sign ambiguity, define

$$D_{\Phi}(z) \equiv \int_0^{\infty} d\mu^2 \frac{\rho_{\Phi}(\mu^2)}{z - \mu^2}, \quad z \in \mathbb{C} \setminus [0, \infty), \quad (158)$$

The non-local part of the Feynman function is the upper-edge boundary value,

$$-i [\tilde{G}_F(s) - P_{\Phi}(s)] = D_{\Phi}(s + i0). \quad (159)$$

The local polynomial has no discontinuity and therefore drops out of every statement below about the cut. For real s in the continuum define the discontinuity by

$$\text{Disc } D_{\Phi}(s) \equiv D_{\Phi}(s + i0) - D_{\Phi}(s - i0). \quad (160)$$

Using

$$\frac{1}{x + i0} - \frac{1}{x - i0} = -2\pi i \delta(x), \quad (161)$$

we obtain

$$\boxed{\text{Disc } D_{\Phi}(s) = -2\pi i \rho_{\Phi}(s), \quad \rho_{\Phi}(s) = -\frac{1}{\pi} \text{Im } D_{\Phi}(s + i0).} \quad (162)$$

Thus the spectral density is the discontinuity of the exact two-point function across its physical cut, with the convention of Eq. (158).

The difference between a stable particle and an unstable resonance is now precise. A stable

particle contributes a delta function to ρ and therefore a pole on the physical sheet. An unstable resonance contributes no isolated delta function. It appears as enhanced weight in the continuum, while the corresponding resonance pole lies on an analytically continued sheet. This is the spectral version of the distinction encountered in the pion electromagnetic form factor.

4.5 The canonical sum rule and why it is not universal

For a canonically normalized elementary scalar field in a regulated theory, the equal-time commutation relation is

$$[\Phi(t, \mathbf{x}), \dot{\Phi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}). \quad (163)$$

The Källén–Lehmann representation implies

$$\langle \Omega | [\Phi(t, \mathbf{x}), \dot{\Phi}(t, \mathbf{y})] | \Omega \rangle = i\delta^{(3)}(\mathbf{x} - \mathbf{y}) \int_0^\infty d\mu^2 \rho_\Phi(\mu^2). \quad (164)$$

Comparing with Eq. (163) gives the sum rule

$$\boxed{\int_0^\infty d\mu^2 \rho_\Phi(\mu^2) = 1.} \quad (165)$$

Since $\rho_\Phi \geq 0$, Eq. (152) then implies

$$0 \leq Z_\Phi^{\text{pole}} \leq 1. \quad (166)$$

This statement has a restricted domain. It relies on the canonical equal-time normalization of the chosen field. A general composite operator need not satisfy Eq. (163), and its spectral density need not obey Eq. (165). Likewise, after a field is rescaled to give unit one-particle residue, its equal-time normalization is correspondingly rescaled. The pole residue and the canonical sum rule are different pieces of information.

This distinction will matter later when we calculate the one-particle overlap of a composite operator such as ϕ^3 .

4.6 Fermionic spectral representation

The same construction applies to a Dirac field, but Lorentz covariance allows two independent scalar spectral functions. Define

$$S_F(x) = \langle \Omega | T \{ \psi(x) \bar{\psi}(0) \} | \Omega \rangle. \quad (167)$$

In a parity-invariant theory its momentum-space representation can be written

$$\boxed{\tilde{S}_F(p) = i \int_0^\infty d\mu^2 \frac{\rho_1(\mu^2) \not{p} + \rho_2(\mu^2)}{p^2 - \mu^2 + i0}.} \quad (168)$$

The two functions are defined by the Lorentz decomposition of the exact fermionic spectral matrix obtained after inserting complete states. Positivity of the Hilbert space implies, for $\mu > 0$,

$$\rho_1(\mu^2) \geq 0, \quad \mu \rho_1(\mu^2) \geq |\rho_2(\mu^2)|, \quad (169)$$

for the parity-invariant decomposition used in Eq. (168).

If the field has one-particle overlap

$$\langle \Omega | \psi(0) | p, s \rangle = (Z_\psi^{\text{pole}})^{1/2} u(p, s), \quad (170)$$

then the isolated one-fermion state contributes

$$\rho_1(\mu^2) \supset Z_\psi^{\text{pole}} \delta(\mu^2 - M^2), \quad \rho_2(\mu^2) \supset M Z_\psi^{\text{pole}} \delta(\mu^2 - M^2). \quad (171)$$

Hence

$$\boxed{\tilde{S}_F(p) \simeq \frac{i Z_\psi^{\text{pole}} (\not{p} + M)}{p^2 - M^2 + i0}} \quad (172)$$

near the physical pole. Once again, M is the pole mass and Z_ψ^{pole} is the exact one-particle pole weight of the chosen fermionic interpolator. For a renormalized composite spinor operator, local polynomial contact terms may again accompany the spectral integral; as in the scalar case they have no isolated one-particle pole and do not alter the pole residue.

4.7 Gauge fields

A covariantly gauge-fixed potential A_μ acts in a state space containing unphysical polarizations, so the simple positivity argument leading to Eq. (140) does not apply to every tensor component of its propagator. The cleanest non-perturbative spectral statements are therefore made either for gauge-invariant operators, such as $F_{\mu\nu}$, or after projection onto the physical transverse sector.

If a physical photon state is present and

$$\langle \Omega | A_\mu(0) | \gamma(p, \lambda) \rangle = (Z_3^{\text{pole}})^{1/2} \epsilon_\mu(p, \lambda), \quad (173)$$

then the transverse part of the two-point function contains a massless one-photon contribution proportional to

$$\frac{i Z_3^{\text{pole}}}{p^2 + i0} \sum_{\lambda=1}^2 \epsilon_\mu(p, \lambda) \epsilon_\nu^*(p, \lambda). \quad (174)$$

The massless case has additional infrared subtleties because the continuum may reach $p^2 = 0$. We shall keep the spectral interpretation but postpone the detailed infrared analysis to the discussion of gauge theories.

4.8 Application: a collider scans a spectral function

The spectral representation becomes especially concrete for a conserved current. Let J_μ^{em} be the hadronic electromagnetic current and define its vacuum polarization tensor by

$$\Pi_{\mu\nu}(q) \equiv i \int d^4x e^{iq \cdot x} \langle \Omega | T \{ J_\mu^{\text{em}}(x) J_\nu^{\text{em}}(0) \} | \Omega \rangle. \quad (175)$$

Current conservation and Lorentz invariance imply

$$\Pi_{\mu\nu}(q) = (q_\mu q_\nu - q^2 \eta_{\mu\nu}) \Pi(q^2), \quad (176)$$

which defines the scalar vacuum-polarization function $\Pi(q^2)$. For timelike $s = q^2$ above hadronic threshold define the electromagnetic spectral function by

$$\rho_{\text{em}}(s) \equiv \frac{1}{\pi} \text{Im} \Pi(s + i0). \quad (177)$$

With the standard normalization

$$J_\mu^{\text{em}} = \sum_f Q_f \bar{q}_f \gamma_\mu q_f, \quad (178)$$

where Q_f is the electric charge of quark flavour f in units of the positron charge e , unitarity gives, to leading order in the electromagnetic coupling while keeping the strong interactions exact,

$$R(s) \equiv \frac{\sigma(e^+e^- \rightarrow \text{hadrons})}{\sigma(e^+e^- \rightarrow \mu^+\mu^-)} = 12\pi \text{Im} \Pi(s + i0), \quad (179)$$

where

$$\sigma(e^+e^- \rightarrow \mu^+\mu^-) = \frac{4\pi\alpha^2}{3s} \quad (180)$$

when the muon mass is neglected, with $\alpha \equiv e^2/(4\pi)$.

Thus an energy scan in e^+e^- annihilation is, in a literal sense, a scan of the spectral content of the electromagnetic current. Thresholds mark the opening of new multiparticle channels. Narrow states produce sharp structures. Broad resonances appear as enhanced weight inside a continuum. At sufficiently high energy the same spectral function can be described usefully in terms of quarks and gluons. The experimental language of “lines”, “thresholds” and “continuum” is the spectral representation made visible.

4.9 Application: dispersion relations and precision observables

Analyticity turns the same spectral information into a relation between real and virtual processes. If one subtraction is sufficient, the vacuum-polarization function obeys

$$\boxed{\Pi(q^2) - \Pi(0) = q^2 \int_{s_0}^{\infty} ds \frac{\rho_{\text{em}}(s)}{s[s - q^2 - i0]}.} \quad (181)$$

The subtraction constant $\Pi(0)$ is separated because the spectral density determines the discontinuity, whereas polynomial terms are invisible to a discontinuity and must be fixed by renormalization conditions.

Equation (181) explains why measured hadronic production can constrain virtual quantum effects. The hadronic contribution to the running electromagnetic coupling and the leading hadronic vacuum-polarization contribution to the anomalous magnetic moment of the muon are obtained from weighted integrals of the same spectral information, with kernels appropriate to the observable. One does not need a weakly coupled description of every hadronic state; unitarity and analyticity allow the measured spectral function to carry the required information.

This is an important modern use of Källén–Lehmann reasoning. A spectral density is not merely a device for proving that propagators have poles and cuts. It is an experimentally accessible object that links inclusive production to virtual corrections.

4.10 Application: Euclidean correlators and lattice spectroscopy

The same spectral logic becomes especially concrete in Euclidean lattice field theory. After the Wick rotation $t = -i\tau$, spacetime is replaced by a lattice with spacing a . Take, for definiteness, a periodic cubic spatial box with N sites in each direction and linear size $L = Na$. At non-zero a and finite L the theory is regulated and the finite-volume spectrum is discrete. Nothing has to be made infinite before masses and operator overlaps can be extracted.

Let $\mathcal{O}(\tau, \mathbf{x})$ be a lattice operator with the desired quantum numbers. Euclidean time is discrete, $\tau = na$ with integer n . Summing over all spatial sites projects the operator to zero spatial momentum,

$$C_{\mathcal{O}}(\tau) = a^3 \sum_{\mathbf{x}} \langle \Omega | \mathcal{O}(\tau, \mathbf{x}) \mathcal{O}^\dagger(0) | \Omega \rangle, \quad \tau = na > 0. \quad (182)$$

Normalize the finite-volume energy eigenstates by $\langle n | m \rangle = \delta_{nm}$. Inserting a complete set gives

$$C_{\mathcal{O}}(\tau) = \sum_n |Z_{\mathcal{O}}^{(n)}(L)|^2 e^{-E_n(L)\tau}, \quad Z_{\mathcal{O}}^{(n)}(L) \equiv \langle \Omega | \mathcal{O}(0) | n, \mathbf{p} = 0; L \rangle. \quad (183)$$

At large Euclidean time the state of lowest energy with non-zero overlap dominates,

$$C_{\mathcal{O}}(\tau) \sim |Z_{\mathcal{O}}^{(1)}(L)|^2 e^{-E_1(L)\tau}. \quad (184)$$

If this level is a stable one-particle state, $E_1(L)$ approaches the infinite-volume pole mass M as $L \rightarrow \infty$, up to the usual finite-volume corrections. The exponent therefore determines the state; the coefficient determines how strongly the chosen operator sees that state. This is the Euclidean counterpart of the distinction between a pole position and an operator-dependent pole residue.

In practice one rarely relies on a single operator. Choose a basis $\{\mathcal{O}_i\}$ of local or spatially smeared interpolators with the same quantum numbers. A smeared interpolator is a weighted average over nearby lattice sites, chosen to improve its overlap with low-lying states without

changing the quantum numbers. Form the correlation matrix

$$C_{ij}(\tau) = a^3 \sum_{\mathbf{x}} \langle \Omega | \mathcal{O}_i(\tau, \mathbf{x}) \mathcal{O}_j^\dagger(0) | \Omega \rangle = \sum_n Z_i^{(n)} Z_j^{(n)*} e^{-E_n \tau}, \quad (185)$$

where $Z_i^{(n)} = \langle \Omega | \mathcal{O}_i | n \rangle$. A non-singular change of interpolating basis,

$$\mathcal{O}_i \longrightarrow \mathcal{O}'_i = R_{ij} \mathcal{O}_j, \quad (186)$$

changes the overlap vectors $Z_i^{(n)}$ and sends $C \rightarrow R C R^\dagger$, but it does not change the energies E_n . The lattice calculation therefore displays, in a particularly direct way, the interpolator-basis independence emphasized throughout these notes: the operator basis is a coordinate choice on the space of interpolators, whereas the spectrum is physical.

The standard variational method makes this separation operational. One solves the generalized eigenvalue problem

$$C(\tau) v^{(n)}(\tau, \tau_0) = \lambda_n(\tau, \tau_0) C(\tau_0) v^{(n)}(\tau, \tau_0), \quad (187)$$

for some reference time τ_0 . When the chosen basis resolves the relevant low-lying states, the principal correlators behave asymptotically as

$$\lambda_n(\tau, \tau_0) \sim e^{-E_n(\tau - \tau_0)}. \quad (188)$$

Changing the interpolating basis changes the eigenvectors and overlaps, but not the extracted energies. This is precisely why lattice calculations deliberately employ many composite operators: a better operator basis improves access to the state without changing the state itself.

There is a second distinction which is equally important. The overlap $Z_i^{(n)} = \langle \Omega | \mathcal{O}_i | n \rangle$ is not the ultraviolet renormalization constant of the lattice operator. To compare a lattice composite operator with a continuum operator defined at a scale μ , one may need a mixing relation of the form

$$\mathcal{O}_a^R(\mu) = Z_{ab}^{\text{UV}}(a, \mu) \mathcal{O}_b^{\text{lat}}, \quad (189)$$

possibly including operators of lower dimension when the symmetries permit. The matrix Z_{ab}^{UV} defines the continuum operator; after that definition is made, its overlap with a particular state is a separate spectral datum. Masses extracted from the exponential fall-off do not require this operator normalization, whereas decay constants and other matrix elements do. This is the lattice version of the distinction that will later appear explicitly for the composite operator ϕ^3 .

Finite volume also turns multiparticle continua into discrete towers of levels. As the volume is enlarged these levels become dense and reconstruct the infinite-volume cuts of the spectral function. Their volume dependence contains scattering information. In the elastic regime, finite-volume quantization conditions of the Lüscher type relate two-particle energy levels to infinite-volume phase shifts, so even the S -matrix can be recovered non-perturbatively from spectral data. A resonance is then identified through the energy dependence of the phase shift (and, after analytic continuation, through its pole), rather than as a new stable one-particle exponential. Once again the interpolating operators determine how efficiently the levels are

created, not the physical scattering amplitude encoded by the levels.

Finally, the lattice viewpoint makes the meaning of renormalization especially transparent. At fixed lattice spacing a every integral and every finite-volume correlator is regulated. Nevertheless one must tune the parameters of the lattice action so that selected physical quantities—masses, decay constants, matrix elements or scattering observables—take prescribed values. One then varies a while holding these quantities fixed, defining a line of constant physics and, when a continuum limit exists, approaching it as $a \rightarrow 0$. Thus the practical act of renormalization is the matching of regulator-dependent parameters and operator normalizations to physical observables. The appearance of ultraviolet divergences in a continuum perturbative expansion is one manifestation of this matching problem, not its definition.

4.11 Application: inclusive response functions

A closely related construction occurs in nuclear and hadronic scattering. Let J_μ be a current coupled to an external probe and $|A\rangle$ a target state. The inclusive hadronic tensor is defined by

$$W_{\mu\nu}(q) = \sum_X (2\pi)^4 \delta^{(4)}(p_A + q - p_X) \langle A | J_\mu(0) | X \rangle \langle X | J_\nu(0) | A \rangle. \quad (190)$$

Here p_A is the target four-momentum and p_X the total four-momentum of the exact final state X . The sum includes every exact final state X accessible to the probe. Equation (190) is a spectral density in the target state rather than in the vacuum. Electron–nucleus and neutrino–nucleus scattering therefore measure response functions built from the same basic object that appears in the Källén–Lehmann argument: a complete sum of squared matrix elements constrained by energy–momentum conservation.

This is why poles, thresholds and continua recur throughout particle and nuclear physics. They are not artifacts of Feynman graphs. They are signatures of the exact spectrum seen through a chosen operator.

4.12 What the non-perturbative part has achieved

We can now state precisely what is known before perturbation theory begins.

A stable particle is an exact one-particle state. A local operator is an interpolator for that particle when its vacuum-to-one-particle matrix element is non-zero. After normalizing this matrix element to the standard free wavefunction, wave packets constructed with the appropriate conserved free inner product generate the asymptotic in- and out-particles. Their overlaps define the S -matrix.

Exact polology then shows that the same particle appears in momentum space as an isolated pole whose residue factorizes into on-shell matrix elements. Sequential extraction of the external poles reproduces the LSZ reduction formula. Finally, the Källén–Lehmann representation places

the isolated pole inside the complete spectral content of the two-point function and defines

$$\begin{aligned} M &= \text{physical pole mass,} \\ Z_{\Phi}^{\text{pole}} &= |\langle \Omega | \Phi(0) | p \rangle|^2 = \text{one-particle pole weight of the chosen operator.} \end{aligned} \tag{191}$$

For a physically normalized interpolating field, the corresponding one-particle weight is unity. For another elementary or composite operator it need not be.

These are exact spectral definitions. The mass M is independent of the choice of interpolating field, whereas Z_{Φ}^{pole} is an exact but operator-dependent spectral weight. Once the external one-particle states have been normalized, the S -matrix is likewise independent of which local interpolators were used to reach them. A physical interaction strength may therefore be defined directly from a scattering amplitude at specified kinematics, or from an analytic continuation of that amplitude when the continuation is free of singularities.

A condition imposed on an off-shell higher-point Green function is different. It can be a very useful way to define a perturbative scheme, but the off-shell vertex can change when the field variables are changed. Part II will keep these two notions separate: physical couplings defined from the S -matrix, and scheme-dependent couplings defined from convenient off-shell Green functions. Neither notion is created by a divergence; the regulator enters only when we calculate their relation to the parameters of a perturbative Lagrangian.

Perturbative realization

Introduction to Part II

Part I has fixed the target. The physical mass is the position of the one-particle pole. The residue tells us how strongly a chosen local operator creates that particle, and LSZ tells us how the corresponding external poles are converted into the S -matrix. These statements were established without expanding in a coupling. We now ask how a perturbative calculation reproduces them.

The answer begins with a regulated Lagrangian and a set of fields chosen for convenience. We calculate self-energies and vertex functions, locate the poles of the resulting Green functions, and impose conditions that match them to the masses and scattering data defined in Part I. The diagrams do not define the physical mass or the normalized particle state; they calculate the relation between those physical quantities and the parameters used in the perturbative Lagrangian.

This is also where a familiar piece of notation becomes subtle. A regulator-dependent field variable—conventionally called a *bare* field—and a new perturbative field variable may be related

by an exact change of variables,

$$\Phi_0 = (Z_\Phi^{\text{ct}})^{1/2} \Phi, \quad (192)$$

where the label “ct” anticipates the counterterm normalization introduced explicitly in Section 6. But this does not mean that an arbitrary perturbative operator is equal, as a complete interacting operator, to $\sqrt{Z_\Phi^{\text{pole}}}$ times the physically normalized interpolating field of Part I. In general,

$$\Phi_{\text{pert}} \neq (Z_\Phi^{\text{pole}})^{1/2} \Phi_R \quad (193)$$

as an operator identity. What LSZ requires is only the corresponding statement for the one-particle projection. Composite operators make this distinction impossible to miss, and we shall use them deliberately.

The same care is needed for couplings. In the perturbative treatment below we shall first impose *mass-shell renormalization* on the elementary fields: the renormalized mass is fixed by the physical pole and the one-particle residue is fixed to unity, with the sign or phase of the one-particle overlap chosen once and for all. For the non-degenerate fields considered here this removes the freedom of linear field redefinition. An off-shell proper vertex evaluated at a specified subtraction point can then be used to define a coupling without any remaining linear field-redefinition ambiguity. Such a coupling can still depend on the subtraction prescription, which is ordinary scheme dependence. A coupling defined directly from an S -matrix element is stronger: it is a physical normalization condition, and its analytic continuation remains independent of the choice of interpolating field. Section 6.8 will make the distinction explicit.

Regularization enters only now. We shall introduce dimensional regularization as a calculational device, derive the counterterm Lagrangian rather than merely write it down, and then carry out mass, field and coupling renormalization explicitly. The same matching would be needed if every regulated loop integral were finite. Divergences tell us how perturbative parameters depend on the regulator; the non-perturbative results of Part I tell us what those parameters are being matched to.

5. Perturbative LSZ and the choice of field representation

Let us begin by translating the exact one-particle pole factorization into diagrams. The local operator used in a perturbative calculation may be elementary or composite. A scalar particle may be interpolated by ϕ , by a renormalized version of ϕ^3 , or by any other local scalar operator with a non-zero vacuum-to-one-particle matrix element. A fermion may likewise be interpolated by ψ or by a composite spinor such as $[\phi\psi]$. Perturbative LSZ must treat all of these choices on the same footing: what matters is the isolated one-particle pole and the operator’s overlap with its residue.

5.1 Local fields and one-particle states

Let $\Phi_A(x)$ be a local operator used in a perturbative calculation. The index A stands for whatever Lorentz indices the operator carries. Suppose that Φ_A has a non-zero overlap with a stable one-particle state $|p, \sigma\rangle$ of mass M . Lorentz covariance fixes the momentum and spin dependence of this matrix element up to an overall constant:

$$\langle \Omega | \Phi_A(0) | p, \sigma \rangle = (Z_\Phi^{\text{pole}})^{1/2} u_A(p, \sigma). \quad (194)$$

Here $u_A(p, \sigma)$ is the standard on-shell wavefunction for the corresponding one-particle representation. We choose the phase of the interpolating operator (or equivalently of the one-particle basis state) so that the coefficient is real and non-negative; without that convention one should write a complex coefficient C_Φ and define $Z_\Phi^{\text{pole}} = |C_\Phi|^2$. With this phase convention Z_Φ^{pole} is defined by Eq. (194). For the moment the superscript “pole” is useful. It reminds us that this Z is the exact one-particle weight of the chosen operator. Unlike the pole mass, it is not representation-independent: rescaling or changing the interpolator changes the weight. It is not a counterterm coefficient: by such a coefficient we shall mean the coefficient of a local term added to the perturbative Lagrangian in order to implement renormalization conditions order by order. Nor is it a renormalization scheme: that term will mean a specified choice of the finite conditions used to define perturbative parameters and field normalizations.

For a scalar,

$$\langle \Omega | \Phi(0) | p \rangle = (Z_\Phi^{\text{pole}})^{1/2}. \quad (195)$$

For a Dirac field,

$$\langle \Omega | \Psi(0) | p, s \rangle = (Z_\Psi^{\text{pole}})^{1/2} u(p, s), \quad \langle \bar{p}, s | \Psi(0) | \Omega \rangle = (Z_\Psi^{\text{pole}})^{1/2} v(p, s), \quad (196)$$

with the corresponding conjugate relations for $\bar{\Psi}$. For a massive vector one has the polarization vector $\epsilon_\mu(p, \lambda)$, and for a photon the same statement is made after projection onto the physical transverse one-photon sector.

The physically normalized interpolating field introduced in Part I obeyed

$$\langle \Omega | \Phi_{R,A}(0) | p, \sigma \rangle = u_A(p, \sigma). \quad (197)$$

Comparing Eqs. (194) and (197) gives a statement about the *one-particle projection* of the two operators. Since fields at a point are operator-valued distributions, the projector statement below is shorthand for the corresponding equality after smearing with a test function, as made explicit later in Section 6. If P_1 denotes the orthogonal projection onto the relevant one-particle subspace, then

$$P_1 \Phi_A(0) | \Omega \rangle = (Z_\Phi^{\text{pole}})^{1/2} P_1 \Phi_{R,A}(0) | \Omega \rangle. \quad (198)$$

This is the statement required by LSZ.

It is worth pausing here because a common textbook shorthand is dangerously imprecise. One

often encounters an equation such as

$$\Phi = Z^{1/2}\Phi_R \tag{199}$$

written as though it were an exact relation between full interacting Heisenberg operators, with the same Z subsequently interpreted as the LSZ residue. In the present meaning of Φ_R , Eq. (199) is generally false. Two local operators may have proportional one-particle projections while having completely different multiparticle matrix elements, short-distance singularities and spectral continua. An elementary operator and a composite operator are the simplest examples. LSZ requires Eq. (198); it does not require Eq. (199).

Later we shall also introduce relations of the form $\Phi_0 = Z_{\text{ct}}^{1/2}\Phi_{\text{ren}}$ as a parametrization of perturbative counterterms. That is a separate use of the symbol Z . In a mass-shell scheme one may choose finite counterterms so that a renormalized perturbative field has unit pole residue, whereas a scheme that subtracts only regulator-dependent singular pieces generally leaves a finite LSZ pole factor. The field counterterm and the pole residue are therefore not the same quantity. Keeping these notions distinct at the outset prevents several familiar confusions.

5.2 Why the scattering matrix is independent of the field representation

The distinction above is not a nuisance. It expresses a basic structural fact about quantum field theory: the physical theory is not the name of a field variable.

In the language of algebraic quantum field theory, the local content of a theory is encoded by a net of local algebras, assigning an algebra to each suitable spacetime region $\mathcal{R}$,

$$\mathcal{R} \longmapsto \mathfrak{A}(\mathcal{R}), \tag{200}$$

represented on a Hilbert space together with the vacuum, the unitary representation of the Poincaré group, and the spectrum of physical states. A particular collection of local fields is a way of generating, or coordinatizing, this local structure. It is not itself the invariant content of the theory. In theories with charged fields one may distinguish a field algebra from the observable algebra, but the same principle remains: physical scattering amplitudes are attached to the asymptotic particle sectors, not to a preferred choice of local generator.

This statement is already implicit in Haag–Ruelle scattering theory. Suppose two local operators Φ and χ carry the same quantum numbers and both have non-zero overlap with the same isolated one-particle mass shell. Normalize their one-particle projections so that

$$\langle \Omega | \Phi_R(0) | p, \sigma \rangle = \langle \Omega | \chi_R(0) | p, \sigma \rangle = u(p, \sigma). \tag{201}$$

The corresponding Haag–Ruelle wave-packet limits construct the same asymptotic one-particle state. Repeating the construction for separated packets gives the same asymptotic Fock space and therefore the same Møller operators. Writing $\Omega_- \equiv \Omega_{\text{in}}$ and $\Omega_+ \equiv \Omega_{\text{out}}$, the scattering

operator

$$S = \Omega_+^\dagger \Omega_- \quad (202)$$

is consequently independent of which normalized interpolator was used to reach the asymptotic particle sector.

Two related statements should not be conflated. The first is *interpolator independence*: two quite different local operators may have non-zero overlap with the same isolated particle and, after their one-particle projections are normalized, construct the same Haag–Ruelle asymptotic states. No invertible map between the operators is required. The pair ϕ and $[\phi^3]_R$ is already of this type.

The second statement is the perturbative *equivalence theorem*. A local, invertible change of Lagrangian field variables, performed order by order in the coupling and leaving the asymptotic particle content unchanged, may alter off-shell Green functions and may reorganize the interaction Lagrangian dramatically. It does not alter on-shell S -matrix elements once the external one-particle states are normalized correctly. In modern algebraic language this is one particular realization of a more general independence from the choice of local generators within the same physical representation: the generators have changed, while the represented local algebra, its particle sectors and the scattering operator have not. Terms proportional to equations of motion are a familiar perturbative manifestation of the equivalence theorem.

Thus interpolator independence is the broader statement; the equivalence theorem is an important special case involving an invertible local reparametrization. There is no preferred “true field” hidden behind perturbation theory. There are physical particle sectors and there are local operators which interpolate them with different efficiencies. The S -matrix belongs to the former.

5.3 Scalar fields: the exact pole and its diagrammatic realization

Before using the diagrammatic language, we fix two terms. A connected graph is called *one-particle irreducible* (1PI) if it cannot be disconnected into two pieces by cutting one internal propagator. To *amputate* a specified external line means to remove the full propagator associated with that line while leaving the remainder of the connected graph unchanged.

We now make the first diagrammatic connection. Let $\Phi(x)$ be a scalar operator, elementary or composite, with

$$\langle \Omega | \Phi(0) | p \rangle = (Z_\Phi^{\text{pole}})^{1/2}. \quad (203)$$

Exact polology and the Källén–Lehmann representation tell us that its two-point function has the form

$$G_{\Phi\Phi}(p) \underset{p^2 \rightarrow M^2}{=} \frac{iZ_\Phi^{\text{pole}}}{p^2 - M^2 + i0} + \text{terms less singular at } p^2 = M^2. \quad (204)$$

Likewise, if p is the momentum carried by one insertion in a connected correlation function,

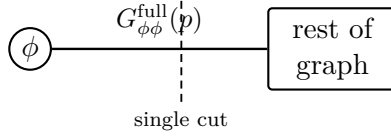
exact polology gives

$$G_{\Phi\chi}(p; \dots) \underset{p^2 \rightarrow M^2}{=} \frac{i(Z_{\Phi}^{\text{pole}})^{1/2}}{p^2 - M^2 + i0} \mathcal{F}(p; \dots) + \text{less singular terms}, \quad (205)$$

where $\mathcal{F}$ is defined only on the mass shell by the exact matrix element obtained after replacing the operator insertion by the physical particle. This is still a non-perturbative statement.

The new ingredient is to ask how the pole in Eq. (205) is produced by perturbative graphs. A simple one-particle pole can arise only from those contributions in which the chosen operator insertion communicates with the rest of the connected graph through a single propagating particle carrying momentum p . If that line is cut, the operator insertion is separated from the remainder of the graph. Contributions which require two or more lines to be cut describe multiparticle intermediate states; they produce thresholds and cuts rather than the isolated simple pole under discussion.

elementary insertion



composite insertion

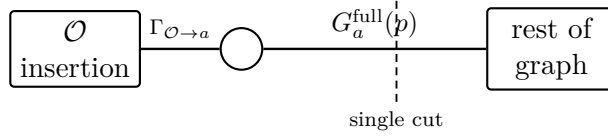


Figure 2: The perturbative realization of an external one-particle pole. For an elementary insertion the full propagator joins the field directly to the rest of the connected graph. A composite insertion may first contain an operator-to-particle subgraph. In both cases the isolated pole is the contribution for which a single physical-particle line separates the insertion from the rest.

For an elementary field ϕ this factorization has the schematic form

$$G_{\phi\chi}(p; \dots) = G_{\phi\phi}(p) \Gamma_{\phi;\chi}(p; \dots) + R(p; \dots), \quad (206)$$

where R contains terms without the one-particle pole in the p channel, and $\Gamma_{\phi;\chi}$ denotes the sum of connected contributions with that external full propagator removed. No assertion is being made that $\Gamma_{\phi;\chi}$ is one-particle irreducible in every channel. It need only be amputated with respect to this particular external particle line.

Using

$$G_{\phi\phi}(p) \sim \frac{iZ_{\phi}^{\text{pole}}}{p^2 - M^2 + i0} \quad (207)$$

and comparing Eq. (206) with exact polology gives, on shell,

$$\mathcal{F}(p; \dots) = (Z_\phi^{\text{pole}})^{1/2} \Gamma_{\phi; \mathcal{X}}(p; \dots) \Big|_{p^2=M^2}. \quad (208)$$

This is the origin of one factor of $\sqrt{Z_\phi^{\text{pole}}}$ for each external scalar leg.

For n external scalar particles, simultaneous factorization of all external one-particle poles gives

$$G_c^{(n)}(p_1, \dots, p_n) \sim \left[\prod_{i=1}^n \frac{i Z_i^{\text{pole}}}{p_i^2 - M_i^2 + i0} \right] \Gamma_{\text{amp}}(p_1, \dots, p_n), \quad (209)$$

where Γ_{amp} is the connected Green function with the external full propagators removed. Exact sequential polology, derived in Section 3, gives instead

$$G_c^{(n)}(p_1, \dots, p_n) \sim \left[\prod_{i=1}^n \frac{i (Z_i^{\text{pole}})^{1/2}}{p_i^2 - M_i^2 + i0} \right] i\mathcal{M}(p_1, \dots, p_n). \quad (210)$$

Comparison of the residues yields the perturbative LSZ rule,

$$i\mathcal{M} = \left[\prod_{i=1}^n (Z_i^{\text{pole}})^{1/2} \right] \Gamma_{\text{amp}} \Big|_{p_i^2=M_i^2}, \quad (211)$$

up to the conventional placement of factors of i in the definition of Γ_{amp} . The content is unambiguous: remove the external full propagators, put the external momenta on their physical mass shells, and multiply by one square root of the exact pole residue of the chosen interpolator for each external particle.

This is the perturbative statement of LSZ. Its diagrammatic ingredient is essential. The exact reduction formula of Part I did not require a graph. Equation (211), by contrast, identifies a particular object inside a diagrammatic expansion—the connected externally amputated function—and relates it to the physical amplitude.

5.4 Composite interpolating operators

The preceding argument becomes more instructive, not less, when the external operator is composite. Let $\mathcal{O}(x)$ be a local scalar composite operator with

$$\langle \Omega | \mathcal{O}(0) | p \rangle = (Z_{\mathcal{O}}^{\text{pole}})^{1/2}. \quad (212)$$

Near the one-particle pole, the perturbative diagrams contributing to a correlation function with an $\mathcal{O}$ insertion have the schematic factorization

$$G_{\mathcal{O}\mathcal{X}}(p; \dots) = \Gamma_{\mathcal{O} \rightarrow a}(p) G_a(p) \Gamma_{a; \mathcal{X}}(p; \dots) + R(p; \dots). \quad (213)$$

Here a labels a field carrying the same one-particle quantum numbers, G_a is its full propagator, and $\Gamma_{\mathcal{O} \rightarrow a}$ is the amputated operator-to-particle insertion. If several fields mix, Eq. (213)

becomes a matrix relation; the principle is unchanged.

The important point is the single cut. The isolated pole comes from the part of the diagram for which one physical-particle line separates the composite insertion from the rest. Everything that happens inside $\Gamma_{\mathcal{O} \rightarrow a}$ belongs to the way the chosen local operator overlaps with that particle. It is not part of the scattering process after the particle has emerged.

If

$$G_a(p) \sim \frac{iZ_a^{\text{pole}}}{p^2 - M^2 + i0}, \quad (214)$$

then comparison with exact polology gives

$$(Z_{\mathcal{O}}^{\text{pole}})^{1/2} = \Gamma_{\mathcal{O} \rightarrow a}(M^2)(Z_a^{\text{pole}})^{1/2}, \quad (215)$$

again up to the normalization convention used for $\Gamma_{\mathcal{O} \rightarrow a}$. This is the diagrammatic definition of the one-particle overlap of a composite interpolator.

Equation (215) should not be confused with ultraviolet renormalization of the composite operator. A composite operator may mix with other local operators of the same quantum numbers and dimension, or with lower-dimensional operators when masses are available. Those mixing coefficients render composite insertions ultraviolet finite. The quantity $Z_{\mathcal{O}}^{\text{pole}}$, by contrast, is the finite exact one-particle pole weight of the resulting operator—the squared magnitude of its overlap with that state. The two notions answer different questions.

This distinction will be tested explicitly in Problem 2 of Assignment 1, with $\mathcal{O} = \phi^3$ in $\lambda\phi^4$ theory. The operator is composite, its one-particle overlap is obtained from the diagrams which leave a single physical scalar line, and no assertion of the form $\phi^3 \propto \phi_R$ as an operator is involved.

5.5 Fermions

Let $\Psi(x)$ be an elementary or composite spinor operator which interpolates a stable fermion of mass M :

$$\langle \Omega | \Psi(0) | p, s \rangle = (Z_{\Psi}^{\text{pole}})^{1/2} u(p, s). \quad (216)$$

Exact polology gives the fermion two-point function near its physical pole,

$$S_{\Psi}(p) \underset{p^2 \rightarrow M^2}{=} \frac{iZ_{\Psi}^{\text{pole}}(\not{p} + M)}{p^2 - M^2 + i0} + \text{less singular terms}. \quad (217)$$

The numerator is not a perturbative decoration; it is the one-particle completeness relation

$$\sum_s u(p, s) \bar{u}(p, s) = \not{p} + M \quad (218)$$

already encountered in the exact theory.

For an elementary perturbative fermion field, a connected Green function with an external

fermion insertion factorizes near the pole as

$$G_\alpha(p; \dots) = S_{\alpha\beta}(p) \Gamma^\beta(p; \dots) + R_\alpha(p; \dots). \quad (219)$$

Comparison with the exact pole residue gives one factor of $(Z_\Psi^{\text{pole}})^{1/2}$ together with the appropriate on-shell spinor. Thus an outgoing fermion contributes

$$(Z_\Psi^{\text{pole}})^{1/2} \bar{u}(p, s), \quad (220)$$

while an incoming fermion contributes

$$(Z_\Psi^{\text{pole}})^{1/2} u(p, s). \quad (221)$$

For antifermions the corresponding factors are v and $\bar{v}$, with the orientation fixed by the usual ordering of spinor indices.

For example, if Γ_{amp} denotes the connected Green function for a process with one incoming and one outgoing fermion after removal of their external full propagators, then schematically

$$i\mathcal{M} = Z_\Psi^{\text{pole}} \bar{u}(p', s') \Gamma_{\text{amp}} u(p, s), \quad (222)$$

with additional $\sqrt{Z^{\text{pole}}}$ factors and wavefunctions for any other external particles.

Nothing in this derivation required Ψ to be elementary. If $\Psi = [\phi\psi]$ or another composite spinor, its operator insertion first factorizes onto the single fermion line exactly as in Eq. (213); the only change is the spinor index carried by the operator-to-particle vertex. The physical amplitude is unchanged after the corresponding one-particle overlap is included.

5.6 Bosons with spin and the photon

The same reasoning applies to any stable bosonic particle. Let Φ_A interpolate a particle of spin j and mass M ,

$$\langle \Omega | \Phi_A(0) | p, \sigma \rangle = (Z_\Phi^{\text{pole}})^{1/2} u_A(p, \sigma). \quad (223)$$

Near the one-particle pole, the two-point function contains the polarization projector

$$G_{AB}(p) \sim \frac{iZ_\Phi^{\text{pole}}}{p^2 - M^2 + i0} \sum_\sigma u_A(p, \sigma) u_B^*(p, \sigma). \quad (224)$$

Diagrammatic factorization removes the external full propagator, and LSZ supplies one factor $\sqrt{Z_\Phi^{\text{pole}}}$ and one physical polarization wavefunction for each external leg.

For a $U(1)$ gauge field the statement must be made on the physical state space. In a fixed covariant gauge one may use A_μ as the perturbative interpolating field, but its Green functions also contain gauge-dependent longitudinal pieces. The physical one-photon matrix element is

$$\langle \Omega | A_\mu(0) | \gamma(p, \lambda) \rangle = (Z_3^{\text{pole}})^{1/2} \epsilon_\mu(p, \lambda), \quad \lambda = 1, 2, \quad (225)$$

with

$$p \cdot \epsilon(p, \lambda) = 0. \quad (226)$$

After removal of the external photon propagator, a physical external photon contributes

$$(Z_3^{\text{pole}})^{1/2} \epsilon_\mu(p, \lambda) \quad (227)$$

or its complex conjugate according to whether the photon is incoming or outgoing. Longitudinal gauge-dependent pieces do not survive the projection onto the physical external state; in perturbative gauge theory this statement is enforced by the Ward identities that will be studied later.

Once again A_μ is not privileged. A gauge-invariant operator such as $F_{\mu\nu}$ also has a one-photon matrix element,

$$\langle \Omega | F_{\mu\nu}(0) | \gamma(p, \lambda) \rangle \propto p_\mu \epsilon_\nu(p, \lambda) - p_\nu \epsilon_\mu(p, \lambda), \quad (228)$$

and may be used as an interpolator. Its off-shell correlators and its pole residue are different, but after normalization it reaches the same physical one-photon state and hence the same scattering matrix.

5.7 What perturbative LSZ actually says

The distinction between the exact and perturbative statements can now be summarized without ambiguity.

The exact reduction theorem of Part I begins with a physically normalized interpolating field and constructs the S -matrix directly from its asymptotic limits. It makes no reference to diagrams. Exact polology then identifies the same reduction as the sequential extraction of one-particle pole residues.

Perturbative LSZ begins instead with whichever local operator—elementary or composite—is convenient for the perturbative calculation. Its one-particle overlap is determined by

$$\langle \Omega | \Phi_A(0) | p, \sigma \rangle = (Z_\Phi^{\text{pole}})^{1/2} u_A(p, \sigma). \quad (229)$$

The diagrammatic expansion is then organized so that the isolated external pole is represented by a single full one-particle propagator joining the operator insertion to the rest of the graph. Removing that propagator gives the connected amputated function, and comparison with exact polology gives the physical amplitude after multiplication by one factor $\sqrt{Z_\Phi^{\text{pole}}}$ and one on-shell wavefunction for every external particle.

The S -matrix is independent of the choice of interpolating operator because different normalized local generators construct the same asymptotic particle sectors. Off-shell Green functions are representation dependent; pole positions and properly normalized on-shell amplitudes are not. This is the content which perturbative field-redefinition invariance must reproduce order by order.

The next task is therefore concrete. We must learn how the perturbative two-point function produces the exact form

$$G_\Phi(p) \underset{p^2 \rightarrow M^2}{\sim} \frac{iZ_\Phi^{\text{pole}}}{p^2 - M^2 + i0}, \quad (230)$$

and how the pole position M and the residue Z_Φ^{pole} are calculated from self-energy diagrams. Here and below the *self-energy* means the 1PI two-point insertion: the part of a two-point graph which cannot be split into two pieces by cutting a single internal propagator. This will make field and mass renormalization operational.

6. Field and mass renormalization: making the exact pole visible order by order

We now turn from the exact pole statement to its perturbative calculation. Let Φ be an elementary or composite local operator which overlaps with a stable particle of mass M . Its exact two-point function has a pole at $p^2 = M^2$, and the residue is the square of its vacuum-to-one-particle matrix element. The pole position belongs to the particle; the residue belongs to the chosen interpolator. For a physically normalized interpolator the residue is one, while for a general perturbative operator it is the Z_Φ^{pole} which appears in LSZ.

The exact theory has already supplied the target for the two-point function: a pole at the physical mass with a residue fixed by the chosen interpolator. Field and mass renormalization are the perturbative procedures by which this pole data are reproduced order by order. For Lagrangian field variables, local counterterms provide a convenient way to impose or match these conditions order by order while the parameters appearing in the Lagrangian are related to physical measurements. For a composite interpolating operator the situation is different: its pole weight is still extracted from the same exact pole theorem, but its ultraviolet definition generally involves operator mixing rather than a kinetic counterterm for the composite operator itself.

By a *counterterm* we mean a local term added to the perturbative Lagrangian, consistent with the symmetries and power counting of the theory, whose coefficient is chosen order by order as part of this matching.

The same matching problem is present even when the loop correction is finite. If such a correction shifted the pole away from the measured mass, the Lagrangian mass parameter would still have to be matched to that measured mass—and, in a mass-shell organization, adjusted so that the two coincide. If a finite momentum-dependent correction changed the pole residue, the normalization of the perturbative field would still have to be adjusted, or the corresponding finite LSZ factor would have to be kept. Ultraviolet divergences make these adjustments unavoidable in most relativistic field theories; they do not define what mass and field renormalization mean.

6.1 Four kinds of data that must be kept distinct

The notation of renormalization is compact, but it can conceal distinctions which are physically important. We shall keep the following objects separate.

Object	Status and exact meaning	Perturbative role
M	Position of the isolated one-particle pole	Fixes the mass counterterm through a two-point renormalization condition
Z_{Φ}^{pole}	Exact but operator-dependent one-particle spectral weight $ \langle\Omega \Phi(0) p\rangle ^2$ (or spinful analogue)	Supplies the external-leg LSZ factor for the chosen elementary or composite interpolator
$\delta Z_{\phi}, \delta Z_{\psi}, \dots$	No independent exact observable meaning	Kinetic counterterms for perturbative Lagrangian field variables; their finite parts depend on the renormalization prescription
$\delta g, \delta\lambda, \dots$	No independent exact observable meaning	Vertex or coupling counterterms fixed by separate conditions on higher-point amplitudes or higher-point vertex functions

A *regulator* is a temporary prescription—for example dimensional regularization, a momentum cutoff or a lattice spacing—which makes otherwise ill-defined intermediate loop expressions well defined. A *renormalization scheme* is a specified set of finite conditions used to define the renormalized parameters and field normalizations once such a prescription has been chosen. In the *mass-shell scheme* used below, the renormalized mass parameter is set equal to the physical pole mass and the elementary perturbative field is normalized to have unit one-particle residue. We also fix the otherwise irrelevant sign or phase of its one-particle overlap. For a single non-degenerate field these conditions fix the linear rescaling freedom completely; with mixing they fix the normalized one-particle basis up to transformations inside an exactly degenerate sector. Consequently, once mass-shell normalization has been imposed, an off-shell vertex condition at a specified momentum point has no remaining ambiguity from the linear field redefinitions admitted in the present perturbative treatment. It may still depend on the chosen subtraction point and is therefore scheme dependent. In a scheme that removes only regulator-dependent singular pieces one normally does not impose the mass-shell residue condition, and the external factor $\sqrt{Z_{\Phi}^{\text{pole}}}$ remains in LSZ. The specific $\overline{\text{MS}}$ and $\overline{\text{MS}}$ prescriptions will be defined after dimensional regularization is introduced in Section 6.5. The counterterm coefficient and the pole residue are therefore related by the chosen renormalization prescription; they are not identical by definition.

The pole mass M is therefore an observable property of the particle sector, whereas Z_{Φ}^{pole} is not

an invariant number attached to the theory by itself. It is an exact datum of the pair “operator Φ + particle state”. This distinction will matter whenever we compare two interpolators for the same state.

The same warning applies, with even greater force, to composite operators. A composite operator may require ultraviolet mixing with other local operators before its correlation functions are finite. After that ultraviolet problem has been solved, it still has a finite one-particle pole weight $Z_{\mathcal{O}}^{\text{pole}}$, equal to the squared magnitude of its one-particle overlap. The two operations answer different questions. Problem 2 of Assignment 1, involving $\mathcal{O} = \phi^3$ in $\lambda\phi^4$ theory, will make this distinction explicit.

6.2 Why a single proportionality constant is all LSZ needs

Suppose Φ is any local scalar interpolator for a stable, non-degenerate particle. A local field is an operator-valued distribution, so the Hilbert-space statement should be made after smearing. Let $h(x)$ be a smooth test function and define

$$\Phi(h) \equiv \int d^4x h(x) \Phi(x), \quad \tilde{h}(p) \equiv \int d^4x e^{ip \cdot x} h(x). \quad (231)$$

If P_1 denotes the orthogonal projector onto the isolated one-particle sector, define the corresponding one-particle packet by

$$|h\rangle_{1p} \equiv \int \frac{d^3p}{(2\pi)^3 2E_p} \tilde{h}(E_p, \mathbf{p}) |p\rangle. \quad (232)$$

Then Lorentz covariance and the matrix element $\langle \Omega | \Phi(0) | p \rangle = C_\Phi$ imply

$$P_1 \Phi(h) | \Omega \rangle = C_\Phi |h\rangle_{1p}. \quad (233)$$

The part orthogonal to this sector contains all other states created by the operator. For a fermion or a particle of higher spin the momentum and polarization dependence is fixed by the corresponding u , v or polarization tensor, and again the remaining freedom for one isolated species is a single scalar coefficient. With the phase convention used above,

$$P_1 \Phi(h) | \Omega \rangle = (Z_\Phi^{\text{pole}})^{1/2} P_1 \Phi_R(h) | \Omega \rangle. \quad (234)$$

A constant rescaling can therefore normalize everything that LSZ sees. If

$$\Phi' = (Z_\Phi^{\text{pole}})^{-1/2} \Phi, \quad (235)$$

then $\langle \Omega | \Phi'(0) | p \rangle = 1$. Nothing has forced the component orthogonal to the one-particle sector to coincide with that produced by some other normalized interpolator. Field normalization calibrates one direction in Hilbert space; it does not identify the full operators.

Källén–Lehmann says the same thing spectrally. If

$$\rho_{\Phi}(s) = Z_{\Phi}^{\text{pole}} \delta(s - M^2) + \rho_{\Phi,c}(s), \quad (236)$$

then after the rescaling above,

$$\rho_{\Phi'}(s) = \delta(s - M^2) + \frac{\rho_{\Phi,c}(s)}{Z_{\Phi}^{\text{pole}}}. \quad (237)$$

The pole has unit weight, but the continuum has not disappeared. This is the cleanest way to see why the useful operation of multiplicative field normalization does *not* imply an exact operator identity of the form $\Phi = \sqrt{Z} \Phi_R$ between independently chosen fields.

There are therefore two different rescalings which textbooks often write with almost identical notation. The exact operation $\Phi \mapsto (Z_{\Phi}^{\text{pole}})^{-1/2} \Phi$ merely turns one chosen interpolator into *its own* unit-pole version. In a regulated perturbative Lagrangian one may instead *define* a new field coordinate by $\phi_0 = (Z_{\phi}^{\text{ct}})^{1/2} \phi$. That equation is an exact relation within the chosen perturbative parametrization. One then chooses the finite part of δZ_{ϕ} so that the field ϕ has unit pole residue order by order. What does *not* follow is either $Z_{\phi}^{\text{ct}} = Z_{\phi}^{\text{pole}}$ as an identity of concepts, or $\phi = (Z_{\phi}^{\text{pole}})^{1/2} \phi_R$ for an independently chosen physically normalized Heisenberg operator. The practical procedure works because the on-shell scattering problem only asks that the chosen perturbative field have the correct normalized one-particle projection; it does not ask that its full operator content coincide with that of ϕ_R .

The same point appears locally in momentum space. A simple pole is characterized by only two Taylor coefficients of the inverse propagator: the position of its zero and the first derivative there. A momentum-dependent self-energy may be arbitrarily complicated away from the pole, but near $p^2 = M^2$,

$$\Pi(p^2) = \Pi(M^2) + (p^2 - M^2)\Pi'(M^2) + \mathcal{O}((p^2 - M^2)^2). \quad (238)$$

Mass renormalization fixes the first number and field normalization fixes the second. Terms quadratic and higher in $(p^2 - M^2)$ change off-shell propagation but neither the position nor the residue of the simple pole. LSZ does not ask us to remove them.

This also explains why a constant kinetic counterterm is sufficient even when the loop self-energy is a non-trivial function of momentum. In a mass-shell organization the local mass and kinetic counterterms subtract the value and the first derivative of the self-energy at the pole. In a different scheme one may leave a finite residue and restore the same scattering amplitude with the explicit LSZ factor. The physics is the matching of the pole data; the counterterm is one convenient technology for implementing that matching. This statement concerns the normalization of a simple one-particle pole. In an effective field theory, higher-derivative local counterterms may still be required to renormalize the full off-shell two-point function order by order; they do not change the fact that only the value and first derivative of the inverse propagator determine the pole mass and residue.

For a composite operator the statement is even more important. The fact that one multiplicative constant is sufficient to normalize its one-particle pole does *not* mean that the composite operator renormalizes multiplicatively in the ultraviolet. An operator such as $[\phi^3]_R$ can require a matrix of UV mixing coefficients while its overlap with a specified one-particle state is still only one number. Problem 2 of Assignment 1 will exhibit precisely this separation.

Finally, one constant is enough only because we have assumed a single isolated, non-degenerate particle species. If several fields interpolate several stable particles with identical quantum numbers, the overlap becomes a matrix,

$$\langle \Omega | \Phi_i(0) | p, a, \sigma \rangle = Z_{ia}^{1/2} u_a(p, \sigma), \quad (239)$$

and field normalization is correspondingly matrix-valued. The general principle is that the normalization data are exactly as large as the one-particle multiplicity space requires.

6.3 The two-point function as the primary renormalization problem

We use the counterterm terminology defined at the beginning of this section and the 1PI notion defined in the preceding section. A *proper n -point vertex* is the amputated sum of 1PI n -point contributions. To avoid a convention clash, we reserve $\Gamma^{(2)}$ below for the inverse two-point function and denote higher proper vertex amplitudes by $\mathcal{V}^{(n)}$ when their explicit Feynman-rule factors of i matter.

For the two-point function, repeated 1PI self-energy insertions build the full propagator. If $G_0(p) = i/(p^2 - m^2 + i0)$ and the self-energy convention is such that an insertion is $-i\Sigma(p^2)$, then

$$\begin{aligned} G(p) &= G_0(p) + G_0(p)[-i\Sigma(p^2)]G_0(p) \\ &\quad + G_0(p)[-i\Sigma(p^2)]G_0(p)[-i\Sigma(p^2)]G_0(p) + \cdots \\ &= \frac{i}{p^2 - m^2 - \Sigma(p^2) + i0}, \end{aligned} \quad (240)$$

with a corresponding change of signs if Σ is defined oppositely. This elementary geometric series is the diagrammatic bridge between a self-energy calculation and the exact pole discussed in Part I.

It is convenient to formulate the rest of the discussion in terms of the exact inverse propagator, whose perturbative expansion is organized by these 1PI two-point insertions. In the sign convention of Eq. (240) the denominator contains $-\Sigma$. To keep the subtraction formulas below visually simple we now define $\Pi_{\text{loop}} \equiv -\Sigma$; no physics is contained in this notational sign. For a scalar perturbative field define

$$G(p) = \frac{i}{\Gamma^{(2)}(p^2) + i0}. \quad (241)$$

At tree level, if the mass parameter appearing in the free Lagrangian is m ,

$$\Gamma_0^{(2)}(p^2) = p^2 - m^2. \quad (242)$$

Loops and local two-point counterterms modify this to

$$\Gamma^{(2)}(p^2) = p^2 - m^2 + \Pi_{\text{loop}}(p^2) + \Pi_{\text{ct}}(p^2), \quad (243)$$

where Π_{loop} denotes the sum of loop contributions in the chosen perturbative organization and Π_{ct} is a polynomial in momentum generated by local counterterms. For a renormalizable scalar theory, the two-point counterterm is at most linear in p^2 ,

$$\Pi_{\text{ct}}(p^2) = A + B(p^2 - m^2). \quad (244)$$

The constant A is the mass counterterm contribution to the inverse propagator; B is the field or kinetic counterterm contribution. This way of writing the counterterm avoids irrelevant sign conventions associated with how one names δm^2 in the Lagrangian. Once a convention for the counterterm Feynman rules is chosen, A and B translate immediately into the usual δm^2 and δZ_ϕ .

Källén–Lehmann and polology require the exact propagator near a stable pole to take the form

$$G(p) = \frac{iZ_\phi^{\text{pole}}}{p^2 - M^2 + i0} + \text{terms regular at } p^2 = M^2. \quad (245)$$

Hence the pole and residue conditions are simply

$$\Gamma^{(2)}(M^2) = 0, \quad \left. \frac{d\Gamma^{(2)}}{dp^2} \right|_{p^2=M^2} = \frac{1}{Z_\phi^{\text{pole}}}. \quad (246)$$

These equations are the precise pole-mass and one-particle-normalization conditions for the chosen perturbative Lagrangian interpolator.

In the mass-shell scheme we choose the perturbative mass parameter itself to be the physical mass,

$$m = M, \quad (247)$$

and we choose the perturbative field to have unit pole residue,

$$Z_\phi^{\text{pole}} = 1. \quad (248)$$

Together with the sign convention $\langle \Omega | \phi(0) | p \rangle > 0$, these conditions fix the linear rescaling freedom of a single non-degenerate scalar field. The pole conditions then reduce to

$$\Gamma^{(2)}(M^2) = 0, \quad \Gamma^{(2)'}(M^2) = 1. \quad (249)$$

Using Eq. (243), the required counterterm polynomial is therefore

$$\Pi_{\text{ct}}^{\text{OS}}(p^2) = -\Pi_{\text{loop}}(M^2) - (p^2 - M^2)\Pi'_{\text{loop}}(M^2). \quad (250)$$

Equivalently, the renormalized loop correction is the twice-subtracted function

$$\Pi_{\text{OS}}(p^2) = \Pi_{\text{loop}}(p^2) - \Pi_{\text{loop}}(M^2) - (p^2 - M^2)\Pi'_{\text{loop}}(M^2), \quad (251)$$

so that

$$\Pi_{\text{OS}}(M^2) = 0, \quad \Pi'_{\text{OS}}(M^2) = 0. \quad (252)$$

This is the cleanest way to see what the two counterterms do. One subtracts the value of the loop correction at the physical pole; the other subtracts its slope there.

6.4 Physical matching at finite cutoff

Suppose the loop contribution is a perfectly finite function near the pole,

$$\Pi_{\text{loop}}(p^2) = a + b(p^2 - M^2) + \mathcal{O}((p^2 - M^2)^2), \quad (253)$$

with finite a and b . If we inserted this expression without adjustment, the propagator would no longer have its pole at the measured mass with unit residue. In a mass-shell counterterm organization one would remove a and b by choosing

$$\Pi_{\text{ct}}^{\text{OS}}(p^2) = -a - b(p^2 - M^2). \quad (254)$$

One could organize the same finite theory differently, but the corresponding finite parameter matching and field normalization would still have to be performed. The same matching conditions apply whether a and b are finite or contain regulator-dependent divergent pieces. The renormalization conditions determine what the parameters mean physically; the divergences determine how the corresponding bare parameters must depend on the regulator.

This point is especially useful when one works with an effective field theory with a physical cutoff, with a lattice regulator, or with a finite ultraviolet completion. There may be no formal infinity in an intermediate expression. The distinction between a Lagrangian parameter and a measured pole mass, and between the normalization of a perturbative field and the residue of its one-particle pole, remains.

In practice the calculation follows a simple order. First choose a regulator so that individual loop expressions are defined. Next write every local counterterm allowed at the order of interest. Then compute the loop part of the inverse propagator, impose the chosen pole and residue conditions, and solve for the two-point counterterms. Coupling, tadpole and composite-operator counterterms are fixed from their own independent conditions. Only after all counterterms required at that order have been inserted consistently should the regulator be removed. The order matters: a divergence tells us that some coefficient depends on the regulator; the renormalization condition tells us which finite physical quantity that coefficient is being used to hold fixed.

6.5 Dimensional regularization

Before evaluating a loop diagram we need a regulator. We shall use dimensional regularization. It preserves Lorentz invariance, allows loop momenta to be shifted without introducing a hard boundary in momentum space, and works particularly well with Feynman parameters.

The prescription is to regard a loop integral as an analytic function of the spacetime dimension d . We write

$$d = 4 - 2\epsilon, \quad (255)$$

evaluate the integral in a region of complex d where it converges, and then analytically continue the result back toward $\epsilon = 0$. An ultraviolet divergence which was present in four dimensions appears as a pole in $1/\epsilon$. The continuation in d is a regulator; it is not a statement that the physical theory lives in a non-integer number of dimensions.

Moving away from four dimensions changes engineering dimensions. For a scalar field,

$$[\phi] = \frac{d-2}{2} = 1 - \epsilon, \quad (256)$$

so a quartic coupling has dimension $4 - d = 2\epsilon$. We introduce an arbitrary mass scale μ_R so that the renormalized coupling may still be written as a dimensionless number, for example $\mu_R^{2\epsilon}\lambda$. The scale μ_R is introduced by the regulator; physical observables cannot depend on it when all orders are included.

Two elementary manipulations will be used repeatedly. First, a Wick rotation sends

$$\ell^0 = i\ell_E^0, \quad \ell^2 = -\ell_E^2, \quad (257)$$

with the Feynman $+i0$ prescription telling us how the contour is rotated.

Feynman's trick: combining several denominators. A loop containing several propagators is awkward because the denominators are centred at different momenta: k , $k+p_1$, $k+p_1+p_2$, and so on. Feynman's parametrization replaces the product by a single denominator. For positive powers α_i define

$$\alpha \equiv \sum_{i=1}^n \alpha_i. \quad (258)$$

Then, with the usual analytic continuation understood,

$$\boxed{\frac{1}{A_1^{\alpha_1} \cdots A_n^{\alpha_n}} = \frac{\Gamma(\alpha)}{\prod_{i=1}^n \Gamma(\alpha_i)} \int_0^\infty \prod_{i=1}^n (dx_i x_i^{\alpha_i-1}) \delta\left(1 - \sum_{i=1}^n x_i\right) \frac{1}{(\sum_{i=1}^n x_i A_i)^\alpha}.} \quad (259)$$

The parameters obey $x_i \geq 0$ and $\sum_i x_i = 1$; geometrically, the integration is over an $(n-1)$ -dimensional simplex. For unit powers this becomes

$$\boxed{\frac{1}{A_1 \cdots A_n} = (n-1)! \int_0^\infty \prod_{i=1}^n dx_i \delta\left(1 - \sum_{i=1}^n x_i\right) \frac{1}{(\sum_{i=1}^n x_i A_i)^n}.} \quad (260)$$

In particular, for two denominators,

$$\boxed{\frac{1}{AB} = \int_0^1 dx \frac{1}{[xA + (1-x)B]^2}}. \quad (261)$$

The real gain is now visible. The linear combination $\sum_i x_i A_i$ is again quadratic in the loop momentum. Completing the square and shifting that momentum leaves a denominator depending only on ℓ^2 and on a function $\Delta(x_i; p_j, m_j)$. Lorentz symmetry is then manifest and the same dimensional-regularization master integral can be used again and again. The proof of the general formula from Schwinger parameters is short but instructive; it is Problem 1(a) of Assignment 1.

A compact derivation of the basic integral uses the Schwinger representation

$$\frac{1}{A^n} = \frac{1}{\Gamma(n)} \int_0^\infty ds s^{n-1} e^{-sA}. \quad (262)$$

After Wick rotation the momentum integral is Gaussian. Rotating back gives

$$\mu_R^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i0)^n} = \frac{i(-1)^n}{(4\pi)^{d/2}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \mu_R^{2\epsilon} (\Delta - i0)^{\frac{d}{2} - n}. \quad (263)$$

The ultraviolet singularity is now contained in the Gamma function. Expanding around $\epsilon = 0$ naturally produces

$$\frac{1}{\epsilon} \equiv \frac{1}{\epsilon} - \gamma_E + \ln 4\pi. \quad (264)$$

We shall need two special cases. With the Feynman prescription understood,

$$I(m) \equiv \mu_R^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2 + i0} = -\frac{m^2}{16\pi^2} \left[\frac{1}{\epsilon} + 1 - \ln \frac{m^2}{\mu_R^2} \right] + \mathcal{O}(\epsilon), \quad (265)$$

and

$$J_2(\Delta) \equiv \mu_R^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i0)^2} = \frac{i}{16\pi^2} \left[\frac{1}{\epsilon} - \ln \frac{\Delta}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \quad (266)$$

The overall sign of a self-energy or vertex contribution still depends on the Feynman-rule convention used to define that object; the pole part of the loop integral itself is fixed by the equations above.

Two subtraction conventions will appear below. In minimal subtraction (MS) one subtracts only the $1/\epsilon$ pole. In modified minimal subtraction ($\overline{\text{MS}}$) one subtracts the combination $1/\bar{\epsilon}$ in Eq. (264). Neither prescription fixes the finite parts by reference to the physical mass shell. Mass-shell renormalization does: its finite two-point counterterms are chosen so that the renormalized masses equal the pole masses and the chosen elementary fields have unit pole residue. Once that linear field normalization has been fixed, higher-point couplings may still be defined at convenient off-shell subtraction points without reintroducing a linear field-redefinition ambiguity.

A useful feature of dimensional regularization is that scaleless integrals vanish. An integral containing neither a mass nor an external momentum has no scale with which to form a non-zero answer, and its ultraviolet and infrared analytic continuations cancel. This fact becomes

especially useful in massless theories and gauge theory.

Nothing in the physical definition of renormalization depends on dimensional regularization. It is simply the device by which the perturbative integrals are made meaningful. The pole mass, the one-particle residue and the S -matrix were defined in Part I before d was moved away from four.

6.6 The regulated and renormalized $\lambda\phi^4$ Lagrangian

We now carry out explicitly the step which is often hidden behind the phrase “add counterterms”. The subscript 0 will denote regulated bare fields and parameters. Include also a vacuum-energy parameter, which will not affect connected flat-space scattering,

$$\mathcal{L}_0 = -\Lambda_0 + \frac{1}{2}(\partial\phi_0)^2 - \frac{1}{2}m_0^2\phi_0^2 - \frac{\lambda_0}{4!}\phi_0^4. \quad (267)$$

The bare quantities are simply the parameters used in the regulated description. At finite cutoff they are ordinary regulator-dependent numbers; if the cutoff is eventually removed, they may have to depend singularly on it.

Introduce a new field variable ϕ by the exact algebraic definition

$$\boxed{\phi_0 = Z_\phi^{1/2}\phi, \quad Z_\phi \equiv 1 + \delta Z_\phi.} \quad (268)$$

Next define the renormalized parameters and their counterterm shifts by

$$\begin{aligned} \Lambda_0 &= \Lambda + \delta\Lambda, \\ Z_\phi(m_0^2 - m^2) &= \delta m^2, \\ Z_\phi^2\lambda_0 &= \mu_R^{4-d}(\lambda + \delta\lambda). \end{aligned} \quad (269)$$

Here μ_R is the reference scale used in dimensional regularization. These equations are definitions of a new set of perturbative coordinates; no approximation and no renormalization condition has yet been imposed.

Substituting Eqs. (268)–(269) into the bare Lagrangian gives, identically,

$$\boxed{\mathcal{L}_0 = \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}},} \quad (270)$$

with

$$\mathcal{L}_{\text{ren}} = -\Lambda + \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\mu_R^{4-d}\lambda}{4!}\phi^4, \quad (271)$$

and

$$\mathcal{L}_{\text{ct}} = -\delta\Lambda + \frac{1}{2}\delta Z_\phi[(\partial\phi)^2 - m^2\phi^2] - \frac{1}{2}\delta m^2\phi^2 - \frac{\mu_R^{4-d}\delta\lambda}{4!}\phi^4. \quad (272)$$

Thus $\mathcal{L}_0$ and $\mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}}$ are not two theories. They are the same regulated theory written with two different sets of variables. The finite parts of m , λ and Z_ϕ acquire meaning only after renormalization conditions are specified.

This is also the place to separate the two uses of a field Z once and for all. Equation (268) is an exact definition inside the perturbative parametrization. It does not assert that the complete interacting operator ϕ is proportional to an independently chosen physically normalized field ϕ_R with proportionality factor $\sqrt{Z_\phi^{\text{pole}}}$. The pole residue is determined by the one-particle projection; Z_ϕ in Eq. (268) is the coefficient used to rewrite the kinetic term. A renormalization condition may relate them, for example by choosing unit pole residue in a mass-shell scheme, but they are not the same object by definition.

The three coefficients play different roles. The two-point conditions determine the mass and kinetic counterterms. A separate condition on a four-point amplitude or proper four-point vertex determines $\delta\lambda$. We introduce this vertex counterterm now because the perturbative theory is not complete without it, but vertex renormalization will be studied systematically later. Nothing in the determination of the pole position and pole residue requires us to solve the four-point problem first.

6.7 One-loop field and mass renormalization in $\lambda\phi^4$ theory

We now use the pole conditions rather than merely quoting them. From $\mathcal{L}_{\text{ren}}$ the free propagator and quartic vertex are

$$\frac{i}{p^2 - m^2 + i0}, \quad -i\mu_R^{2\epsilon}\lambda. \quad (273)$$

The quadratic counterterms in Eq. (272) give a two-point insertion

$$i \left[\delta Z_\phi(p^2 - m^2) - \delta m^2 \right]. \quad (274)$$

At one loop the only 1PI two-point loop graph is the tadpole. The three objects which enter the renormalized inverse propagator at this order are therefore the tadpole, the mass-counterterm insertion and the kinetic-counterterm insertion.

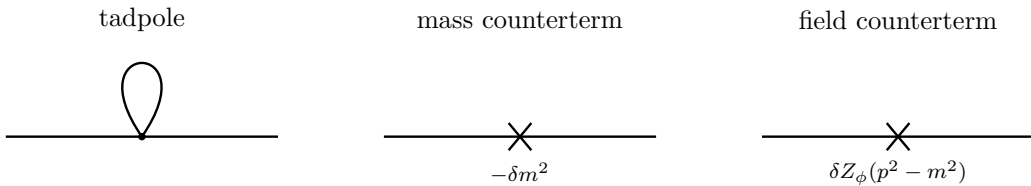


Figure 3: The complete one-loop two-point renormalization problem in $\lambda\phi^4$ theory. The tadpole is momentum independent. The two local counterterms are exactly the structures needed to adjust the value and the slope of the inverse propagator at the physical pole.

At one loop the only one-particle-irreducible two-point graph in $\lambda\phi^4$ theory is the tadpole. The first momentum-dependent self-energy appears at two loops in the sunset graph.

With the sign convention of Eq. (243), the one-loop tadpole contribution to the inverse propagator is

$$\Pi_{\text{loop}}^{(1)}(p^2) = -\frac{\lambda}{2}I(m) = \frac{\lambda m^2}{32\pi^2} \left[\frac{1}{\epsilon} + 1 - \ln \frac{m^2}{\mu_R^2} \right] + \mathcal{O}(\epsilon), \quad (275)$$

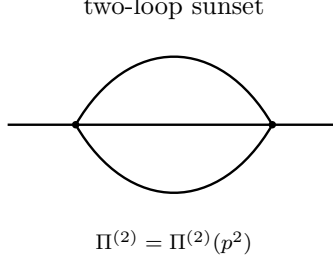


Figure 4: The first momentum-dependent self-energy in $\lambda\phi^4$ theory is the two-loop sunset. Field renormalization therefore begins one loop later than mass renormalization in this theory.

where Eq. (265) has been used. The important point is that the answer contains no external momentum. At one loop the complete inverse propagator is therefore

$$\Gamma^{(2)}(p^2) = p^2 - m^2 + \Pi_{\text{loop}}^{(1)} + \delta Z_{\phi}^{(1)}(p^2 - m^2) - \delta m_{(1)}^2 + \mathcal{O}(\lambda^2). \quad (276)$$

This equation contains the whole one-loop mass-and-field renormalization problem.

In the mass-shell scheme we set $m = M$ and impose

$$\Gamma^{(2)}(M^2) = 0, \quad \Gamma^{(2)'}(M^2) = 1. \quad (277)$$

The first condition gives

$$\delta m_{(1),\text{OS}}^2 = \Pi_{\text{loop}}^{(1)}(M^2) = \frac{\lambda M^2}{32\pi^2} \left[\frac{1}{\bar{\epsilon}} + 1 - \ln \frac{M^2}{\mu_R^2} \right]. \quad (278)$$

The second condition gives

$$\delta Z_{\phi}^{(1)} = -\Pi_{\text{loop}}^{(1)'}(M^2) = 0. \quad (279)$$

Thus the one-loop divergence is removed entirely by the mass counterterm. There is no one-loop field counterterm because there is no momentum dependence to subtract.

The same calculation looks slightly different in $\overline{\text{MS}}$. There we remove only the $1/\bar{\epsilon}$ part,

$$\delta m_{(1),\overline{\text{MS}}}^2 = \frac{\lambda m^2}{32\pi^2} \frac{1}{\bar{\epsilon}}, \quad \delta Z_{\phi,\overline{\text{MS}}}^{(1)} = 0, \quad (280)$$

and leave the finite difference between m and the pole mass to be determined from the pole equation. The physical answer is the same; the bookkeeping is different.

This example separates mass and field renormalization in the cleanest possible way. The value of the self-energy at the pole changes the pole position. The derivative of the self-energy changes the residue. A tadpole has a value but no slope. Hence mass renormalization occurs at one loop, while field renormalization first appears at two loops through the sunset graph in Fig. 4.

The exact Källén–Lehmann form makes the same point from the other side,

$$G_\phi(p) \sim \frac{iZ_\phi^{\text{pole}}}{p^2 - M^2 + i0}. \quad (281)$$

The perturbative calculation above has simply reproduced the first two pieces of exact pole data: its position and its residue.

6.8 One-loop four-point function and coupling renormalization

The two-point function cannot determine $\delta\lambda$. To fix the quartic coupling we must look at a four-point quantity. For the process $p_1 + p_2 \rightarrow p_3 + p_4$ define the Mandelstam invariants

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2. \quad (282)$$

They are the invariant momentum squares carried by the three possible pairings of the external legs. At tree level the proper four-point Feynman amplitude is simply $-i\lambda$ in four dimensions. At one loop there are three bubble diagrams, corresponding to the s , t and u channels, together with the local $\delta\lambda$ counterterm.

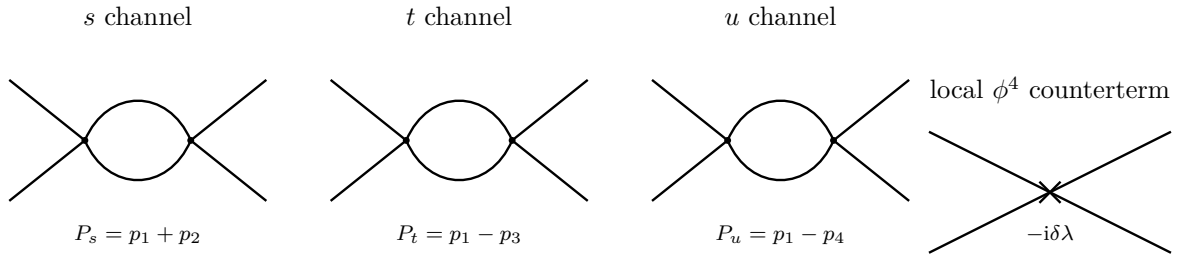


Figure 5: The three one-loop bubble contributions to the quartic vertex and the local counterterm. The graph is the same topology in each channel; what changes is the momentum flowing through the bubble.

For the s channel define $P^2 = s$. This is the first place where we shall use Feynman’s trick in detail. Write

$$D_1 = k^2 - m^2 + i0, \quad D_2 = (k + P)^2 - m^2 + i0. \quad (283)$$

The two-denominator formula gives

$$\frac{1}{D_1 D_2} = \int_0^1 dx \frac{1}{[xD_1 + (1-x)D_2]^2}. \quad (284)$$

The important step is to complete the square rather than simply quote the result:

$$\begin{aligned} xD_1 + (1-x)D_2 &= x(k^2 - m^2) + (1-x)[(k+P)^2 - m^2] + i0 \\ &= k^2 + 2(1-x)k \cdot P + (1-x)P^2 - m^2 + i0 \\ &= [k + (1-x)P]^2 - [m^2 - x(1-x)P^2] + i0. \end{aligned} \quad (285)$$

Thus with

$$\ell = k + (1-x)P, \quad \Delta(x, s) = m^2 - x(1-x)s, \quad (286)$$

the product of the two propagators has become a single rotationally invariant denominator. Including the two factors of i from the propagators,

$$\begin{aligned} B(s) &\equiv \mu_R^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{i}{D_1} \frac{i}{D_2} \\ &= -\mu_R^{2\epsilon} \int_0^1 dx \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{[\ell^2 - \Delta(x, s) + i0]^2} \\ &= -\frac{i}{16\pi^2} \left[\frac{1}{\epsilon} - \int_0^1 dx \ln \frac{m^2 - x(1-x)s - i0}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \end{aligned} \quad (287)$$

The last line is now just the master integral in Eq. (266). Notice something that will recur throughout renormalization: the ultraviolet pole is independent of s . The divergent part is therefore local and can be cancelled by the momentum-independent ϕ^4 counterterm. Assignment 1, Problem 1 asks the reader to use the same method in a different renormalizable theory, where both two- and three-denominator integrals occur. The t - and u -channel expressions are obtained by replacing s by t and u . Including the symmetry factor $1/2$ for each bubble, the one-loop proper four-point amplitude is

$$\mathcal{V}^{(4)}(s, t, u) = -i\lambda + \frac{(-i\lambda)^2}{2} [B(s) + B(t) + B(u)] - i\delta\lambda + \mathcal{O}(\lambda^3). \quad (288)$$

Its ultraviolet part is local and independent of the external kinematics,

$$\mathcal{V}^{(4)}|_{\text{div}} = i \frac{3\lambda^2}{32\pi^2} \frac{1}{\epsilon} - i\delta\lambda|_{\text{div}}. \quad (289)$$

Thus in $d = 4 - 2\epsilon$ the $\overline{\text{MS}}$ counterterm is

$$\boxed{\delta\lambda_{\overline{\text{MS}}}^{(1)} = \frac{3\lambda^2}{32\pi^2} \frac{1}{\epsilon}}. \quad (290)$$

The ultraviolet pole fixes only the divergent part of $\delta\lambda$. Its finite part depends on how we choose to define the renormalized coupling. Here an important distinction has to be made.

A common momentum-subtraction (MOM) convention uses an off-shell *Euclidean symmetric point*. Take all four external momenta to be incoming, continue them to Euclidean vectors k_i , and impose

$$\sum_{i=1}^4 k_i = 0, \quad k_i^2 = Q^2, \quad k_i \cdot k_j = -\frac{Q^2}{3} \quad (i \neq j). \quad (291)$$

After continuation back to Minkowski invariants the external legs are spacelike, $p_i^2 = -Q^2$, and

$$s = t = u = -\frac{4Q^2}{3}. \quad (292)$$

No channel momentum vanishes and the point lies away from physical thresholds. A MOM

coupling may therefore be defined by

$$\mathcal{V}_R^{(4)}|_{\mathcal{K}_*(Q)} = -i\lambda_{\text{MOM}}(Q). \quad (293)$$

At first sight an off-shell condition such as Eq. (293) might seem to reintroduce the field-redefinition ambiguity that Part I taught us to avoid. In the present scheme it does not. Before defining the vertex we have already imposed mass-shell normalization on the elementary scalar field: its pole is at M^2 , its residue is one, and the sign of its one-particle overlap has been fixed. For the single non-degenerate scalar considered here these conditions remove the freedom of linear field redefinition. Therefore the value of the off-shell proper vertex at the specified Euclidean symmetric point is unambiguous under the class of linear field redefinitions used in this perturbative treatment. In this precise sense, *mass-shell renormalization fixes the field-redefinition ambiguity of the off-shell vertex normalization*.

The resulting number $\lambda_{\text{MOM}}(Q)$ is nevertheless scheme dependent: changing Q or choosing another subtraction configuration gives another finite definition of the coupling. Scheme dependence is not the same thing as field-redefinition ambiguity. Once both the mass-shell normalization of the fields and the off-shell subtraction prescription have been specified, the perturbative coupling is completely defined.

There is a more general statement which is useful to keep in mind but which we shall not need in the calculations below. If one allows nonlinear local redefinitions of the interacting field, off-shell Green functions and proper vertices can change even after the one-particle pole has been normalized. Such transformations lie outside the linear field parametrization used in this part of the course. The physical S -matrix, however, remains unchanged.

For comparison, the LSZ-normalized scattering amplitude is field-redefinition independent without this restriction. For identical stable scalars the on-shell amplitude $\mathcal{M}(s, t, u)$ obeys $s + t + u = 4M^2$ and, away from poles and cuts, is the boundary value of an analytic function of the Mandelstam variables. Equality of the physical S -matrix in two field descriptions therefore continues into any common analytic domain that can be reached without crossing a singularity. One may consequently define a representation-independent coupling from an analytically continued amplitude. For equal masses a natural subthreshold crossing-symmetric point is

$$s = t = u = \frac{4M^2}{3}, \quad \mathcal{M}\left(\frac{4M^2}{3}, \frac{4M^2}{3}, \frac{4M^2}{3}\right) \equiv -\lambda_{\text{phys}}. \quad (294)$$

This is not the Euclidean symmetric point of Eq. (291): here the external particles remain on their mass shells and only the Mandelstam invariants are analytically continued, whereas at the Euclidean symmetric point the external legs themselves are off shell.

We shall use the first construction in the perturbative examples: mass-shell normalization fixes the linear field variables, and a specified Euclidean symmetric point then fixes the vertex coupling. The second construction explains how the same interaction can, when desired, be normalized directly by a field-redefinition-independent scattering quantity.

The separation between two-point and four-point renormalization is now concrete. The pole

conditions determined δm^2 and δZ_ϕ . They could not determine $\delta\lambda$, because the quartic local structure is independent information. The four-point sector supplies the missing condition, either as a chosen off-shell scheme such as MOM or as a physical normalization of the scattering amplitude.

6.9 Example II: a fermion coupled to a scalar

We now repeat the same exercise for Yukawa theory. There is one extra point which should be made before writing the Lagrangian. With no additional symmetry that forbids odd powers of a real scalar, quantum corrections can generate both a one-point function and a cubic scalar interaction. We therefore denote by t_0 the bare coefficient of the linear term $-t_0\phi_0$ and by κ_0 the bare cubic coupling. The parameter t is usually called the *tadpole parameter*. If we choose the vacuum so that $\langle\phi\rangle = 0$, its renormalized value is fixed by requiring the renormalized one-point proper function to vanish,

$$\Gamma_\phi^{(1)} = 0. \quad (295)$$

If an additional discrete symmetry forbids odd scalar operators, t , κ and their counterterms can of course be omitted consistently.

The most general regulated renormalizable Lagrangian needed for the present discussion is

$$\begin{aligned} \mathcal{L}_0 = & -\Lambda_0 + \frac{1}{2}(\partial\phi_0)^2 - t_0\phi_0 - \frac{1}{2}\mu_0^2\phi_0^2 - \frac{\kappa_0}{3!}\phi_0^3 - \frac{\lambda_0}{4!}\phi_0^4 \\ & + \bar{\psi}_0(i\not{\partial} - m_0)\psi_0 - g_0\phi_0\bar{\psi}_0\psi_0. \end{aligned} \quad (296)$$

Define new perturbative field variables by

$$\phi_0 = Z_\phi^{1/2}\phi, \quad \psi_0 = Z_\psi^{1/2}\psi, \quad Z_\phi = 1 + \delta Z_\phi, \quad Z_\psi = 1 + \delta Z_\psi, \quad (297)$$

and define the parameter shifts by

$$\begin{aligned} \Lambda_0 &= \Lambda + \delta\Lambda, \\ Z_\phi^{1/2}t_0 &= t + \delta t, \\ Z_\phi(\mu_0^2 - \mu^2) &= \delta\mu^2, \\ Z_\phi^{3/2}\kappa_0 &= \kappa + \delta\kappa, \\ Z_\phi^2\lambda_0 &= \mu_R^{4-d}(\lambda + \delta\lambda), \\ Z_\psi(m_0 - m) &= \delta m, \\ Z_\psi Z_\phi^{1/2}g_0 &= \mu_R^{(4-d)/2}(g + \delta g). \end{aligned} \quad (298)$$

The dimensionful parameters are understood with their d -dimensional engineering dimensions; the displayed powers of μ_R keep the couplings which are dimensionless in four dimensions dimensionless away from $d = 4$.

Substitution now gives the exact identity

$$\boxed{\mathcal{L}_0 = \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}},} \quad (299)$$

where

$$\begin{aligned} \mathcal{L}_{\text{ren}} = & -\Lambda + \frac{1}{2}(\partial\phi)^2 - t\phi - \frac{1}{2}\mu^2\phi^2 - \frac{\kappa}{3!}\phi^3 - \frac{\mu_R^{4-d}\lambda}{4!}\phi^4 \\ & + \bar{\psi}(i\not{\partial} - m)\psi - \mu_R^{(4-d)/2}g\phi\bar{\psi}\psi, \end{aligned} \quad (300)$$

and

$$\begin{aligned} \mathcal{L}_{\text{ct}} = & -\delta\Lambda + \frac{1}{2}\delta Z_\phi[(\partial\phi)^2 - \mu^2\phi^2] - \delta t\phi - \frac{1}{2}\delta\mu^2\phi^2 - \frac{\delta\kappa}{3!}\phi^3 - \frac{\mu_R^{4-d}\delta\lambda}{4!}\phi^4 \\ & + \delta Z_\psi\bar{\psi}(i\not{\partial} - m)\psi - \delta m\bar{\psi}\psi - \mu_R^{(4-d)/2}\delta g\phi\bar{\psi}\psi. \end{aligned} \quad (301)$$

Again the split is algebra, not an additional physical assumption. The various renormalization conditions tell us how the finite parts of these parameters are to be tied to observables. The one-point condition fixes t or δt once a vacuum convention is chosen; the scalar and fermion two-point functions fix their masses and pole normalizations; higher-point amplitudes fix g , κ and λ according to the chosen scheme. For the fermion pole studied next, only m and δZ_ψ enter directly at one loop, while δg belongs to the three-point sector and will be treated systematically later.

6.10 The exact fermion pole dictates the renormalization conditions

For the massive case considered here, $m \neq 0$. Lorentz invariance and parity imply that the exact fermion inverse propagator can be written as

$$\Gamma_\psi^{(2)}(p) = A(p^2)\not{p} - B(p^2), \quad (302)$$

where A and B are scalar functions. Equivalently, if one starts from a perturbative mass parameter m and defines the loop self-energy by

$$\Sigma(\not{p}) = \not{p}\Sigma_V(p^2) + m\Sigma_S(p^2), \quad (303)$$

then

$$A(p^2) = 1 - \Sigma_V(p^2) + \text{counterterms}, \quad B(p^2) = m[1 + \Sigma_S(p^2)] + \text{counterterms}. \quad (304)$$

No other Dirac structure is available in this parity-preserving problem.

The physical mass M is the point at which the inverse propagator annihilates an on-shell spinor,

$$\Gamma_\psi^{(2)}(p)u(p, s) = 0, \quad p^2 = M^2, \quad (\not{p} - M)u(p, s) = 0. \quad (305)$$

Equation (302) therefore gives

$$B(M^2) = MA(M^2). \quad (306)$$

This is the fermionic form of the statement that the physical mass is the pole position.

Near the pole the propagator must have the exact polological form

$$S(p) = \frac{iZ_\psi^{\text{pole}}(\not{p} + M)}{p^2 - M^2 + i0} + \text{terms regular at } p^2 = M^2. \quad (307)$$

Expanding Eq. (302) around $p^2 = M^2$ gives

$$(Z_\psi^{\text{pole}})^{-1} = A(M^2) + 2M^2 A'(M^2) - 2MB'(M^2). \quad (308)$$

Thus the two-point counterterms are again fixed by two independent pieces of pole data: the location of the zero of the inverse propagator and its first variation around that zero.

If the perturbative mass is chosen to be the physical mass, $m = M$, and the field is normalized on shell so that $Z_\psi^{\text{pole}} = 1$, the renormalized self-energy functions obey

$$\Sigma_V^R(M^2) + \Sigma_S^R(M^2) = 0, \quad (309)$$

while the residue condition gives

$$\Sigma_V^R(M^2) + 2M^2 [\Sigma_V^{R'}(M^2) + \Sigma_S^{R'}(M^2)] = 0. \quad (310)$$

These two equations determine the finite as well as divergent parts of δm and δZ_ψ in the mass-shell scheme.

6.11 The one-loop Yukawa self-energy: enough calculation to see the answer

At one loop the fermion emits and reabsorbs a scalar.

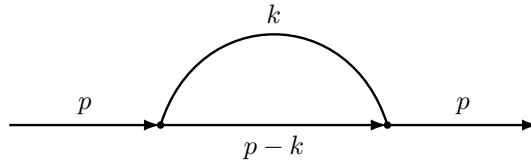


Figure 6: The one-loop fermion self-energy in Yukawa theory. The scalar and fermion propagators share the external momentum, so the loop correction is already a non-trivial function of p^2 at order g^2 .

With the convention $S^{-1}(p) = \not{p} - m - \Sigma(\not{p})$, the one-loop insertion is

$$-i\Sigma^{(1)}(\not{p}) = (-i\mu_R^{(4-d)/2}g)^2 \int \frac{d^d k}{(2\pi)^d} \frac{i(\not{p} - \not{k} + m)}{(p - k)^2 - m^2 + i0} \frac{i}{k^2 - \mu^2 + i0}. \quad (311)$$

Introduce a Feynman parameter and shift $k = \ell + xp$. The denominator becomes

$$[\ell^2 - \Delta(x, p^2) + i0]^2, \quad \Delta(x, p^2) = xm^2 + (1-x)\mu^2 - x(1-x)p^2, \quad (312)$$

while the numerator becomes $(1-x)\not{p} - \not{\ell} + m$. The odd term in ℓ integrates to zero. For $d = 4 - 2\epsilon$ we use

$$\mu_R^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i0)^2} = \frac{i}{16\pi^2} \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \quad (313)$$

Hence, with the sign convention of Eq. (311),

$$\boxed{\Sigma^{(1)}(\not{p}) = -\frac{g^2}{16\pi^2} \int_0^1 dx [(1-x)\not{p} + m] \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta(x, p^2)}{\mu_R^2} \right]}. \quad (314)$$

We can now read off, rather than merely infer, the two scalar functions,

$$\begin{aligned} \Sigma_V^{(1)}(p^2) &= -\frac{g^2}{16\pi^2} \int_0^1 dx (1-x) \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta(x, p^2)}{\mu_R^2} \right], \\ \Sigma_S^{(1)}(p^2) &= -\frac{g^2}{16\pi^2} \int_0^1 dx \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta(x, p^2)}{\mu_R^2} \right]. \end{aligned} \quad (315)$$

Their divergent parts are therefore

$$\Sigma_V^{(1)} \Big|_{\text{div}} = -\frac{g^2}{32\pi^2} \frac{1}{\bar{\epsilon}}, \quad \Sigma_S^{(1)} \Big|_{\text{div}} = -\frac{g^2}{16\pi^2} \frac{1}{\bar{\epsilon}}. \quad (316)$$

The pole equation derived in the previous subsection immediately gives, to order g^2 ,

$$M - m = m \left[\Sigma_V^{(1)}(m^2) + \Sigma_S^{(1)}(m^2) \right] + \mathcal{O}(g^4). \quad (317)$$

Thus

$$M - m = -\frac{g^2 m}{16\pi^2} \int_0^1 dx (2-x) \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta(x, m^2)}{\mu_R^2} \right] + \mathcal{O}(g^4), \quad (318)$$

so the mass counterterm must cancel the ultraviolet pole proportional to $3g^2 m/(32\pi^2 \bar{\epsilon})$, with its sign fixed by the counterterm convention.

The residue formula is equally explicit. To one loop we may put $m = M$ inside terms already proportional to g^2 , and Eq. (308) gives

$$\begin{aligned} (Z_\psi^{\text{pole}})^{-1} &= 1 + \frac{g^2}{16\pi^2} \int_0^1 dx (1-x) \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta(x, M^2)}{\mu_R^2} \right] \\ &\quad + \frac{g^2}{8\pi^2} M^2 \int_0^1 dx \frac{x(1-x)(2-x)}{\Delta(x, M^2)} + \mathcal{O}(g^4). \end{aligned} \quad (319)$$

Only the first integral contains an ultraviolet pole; the derivative terms are ultraviolet finite. The kinetic counterterm cancels that pole, and in a mass-shell scheme its finite part is chosen so

that the final renormalized perturbative field has unit pole residue.

Three facts are now explicit. First, the term proportional to m shifts the pole. Second, the term proportional to $\not{p}$ changes the normalization. Third, the logarithm of $\Delta(x, p^2)$ makes both functions genuinely momentum dependent. Thus both mass and field renormalization are already non-trivial at order g^2 .

The contrast with one-loop $\lambda\phi^4$ theory is immediate. There the first self-energy is momentum independent and only the pole moves. Here the first self-energy is already momentum dependent, so both the pole and its residue move.

The scalar propagator in Yukawa theory gives the same lesson from the bosonic side. A fermion loop contributes to the scalar two-point function with non-trivial dependence on the external momentum. Consequently the scalar kinetic counterterm δZ_ϕ is already required at one loop in the Yukawa theory. We do not need its explicit integral here; the exact pole theorem tells us what pieces of that integral must eventually be extracted.

6.12 The role of the Yukawa vertex counterterm

The counterterm δg in Eq. (301) cannot be fixed from either the fermion or scalar propagator alone. It belongs to the three-point sector. In the mass-shell scheme the scalar and fermion masses and one-particle residues have already been fixed, together with the phase conventions for their one-particle states. Within the linear field parametrization used here, this removes the field-redefinition ambiguity from a subsequent off-shell momentum-subtraction condition on the proper $\phi\bar{\psi}\psi$ vertex. The resulting g is still scheme dependent because it depends on the chosen subtraction kinematics, but it is unambiguous once those kinematics and the mass-shell field normalization have been specified. Alternatively, whenever a suitable decay or scattering process exists, one may define a physical Yukawa coupling from an LSZ-normalized amplitude; an analytic continuation of that amplitude is field-redefinition independent without reference to an off-shell field convention.

Once a condition has fixed δg , its counterterm insertions must be included consistently in higher-order two-point calculations. This is all that is needed about vertex renormalization at present. The systematic study of proper vertices, subdivergences and coupling renormalization belongs later. The important point is that two-point and higher-point conditions fix independent local structures, even though their counterterms begin to appear together inside higher-order diagrams.

6.13 Mass-shell normalization, minimal subtraction and the meaning of Z

The mass-shell scheme is particularly transparent for the present discussion because it builds the exact pole theorem directly into the perturbative field and, for the non-degenerate fields treated here, fixes the linear field-redefinition freedom before any off-shell vertex normal-

ization is imposed. For a scalar,

$$\Gamma^{(2)}(M^2) = 0, \quad \Gamma^{(2)'}(M^2) = 1, \quad (320)$$

and for a fermion Eqs. (309) and (310) impose the corresponding conditions. The renormalized perturbative field then has

$$Z_{\Phi}^{\text{pole}} = 1, \quad (321)$$

so the external LSZ factor associated with that field is trivial.

$\overline{\text{MS}}$ and $\overline{\text{MS}}$ make a different choice. They remove only the regulator-dependent pole terms defined in Section 6.5 and do not normally adjust finite parts to place the propagator in unit-residue form at the physical pole. The renormalized mass parameter is then not automatically the physical pole mass, and the renormalized field does not automatically have $Z_{\Phi}^{\text{pole}} = 1$. The physical mass is obtained by solving the pole equation, and perturbative LSZ still supplies the finite external factor $\sqrt{Z_{\Phi}^{\text{pole}}}$.

Neither prescription changes the S -matrix. They are different choices of perturbative coordinates and finite normalization conditions. In the mass-shell prescription used for our vertex examples, the linear field normalization is fixed first; the subsequent off-shell vertex condition therefore introduces scheme dependence but no further linear field-redefinition ambiguity.

It is therefore useful to reserve the notation

$$Z_{\Phi}^{\text{pole}} \quad (322)$$

for the spectral or LSZ residue, and to use

$$\delta Z_{\Phi} \quad \text{or} \quad Z_{\Phi}^{\text{ct}} = 1 + \delta Z_{\Phi} \quad (323)$$

for the counterterm normalization of a perturbative field variable. They can be related by a renormalization condition; they should not be identified before that condition is stated.

6.14 A preview: what changes for a composite operator such as ϕ^3

Problem 2 of Assignment 1 will revisit the same structure for $\mathcal{O} = \phi^3$ in $\lambda\phi^4$ theory, where the distinction becomes especially transparent. The composite operator can interpolate the same one-particle scalar state as ϕ because

$$\langle \Omega | \phi^3(0) | p \rangle \neq 0 \quad (324)$$

in the interacting theory. Its exact mixed correlator therefore contains the same physical pole $p^2 = M^2$. What changes is the residue and the continuum.

Diagrammatically, the one-particle pole is obtained from those contributions in which the composite insertion is connected to the rest of the graph by a single physical scalar line. The coefficient of that pole defines $Z_{\phi^3}^{\text{pole}}$. At the same time the local operator ϕ^3 has its own

ultraviolet renormalization problem and can mix with other operators of the same quantum numbers, including lower-dimensional operators when powers of the mass are available. Those mixing coefficients are not $Z_{\phi^3}^{\text{pole}}$.

The exact theorem tells us what survives all these complications. After the operator has been properly defined, its one-particle projection is still just a number times the physically normalized one-particle state. That one number is all perturbative LSZ needs. The full operator need not, and generally does not, become proportional to ϕ_R .

6.15 What the non-perturbative theory has taught perturbation theory

We can now summarize the logic of mass and field renormalization in the order in which the physics determines it:

$$\begin{aligned} \text{exact spectrum} &\longrightarrow \text{pole position and residue} \longrightarrow \text{renormalization conditions}, \\ \text{renormalization conditions} &\longrightarrow \text{counterterms} \longrightarrow \text{finite-order prediction}. \end{aligned} \tag{325}$$

The first two arrows are non-perturbative. They were established by scattering theory, polology and the Källén–Lehmann representation. The last two arrows are perturbative technology.

For $\lambda\phi^4$ theory the one-loop tadpole moves the pole but does not alter its slope, so mass renormalization appears one loop before field renormalization. For a fermion in Yukawa theory the one-loop self-energy is already momentum dependent, so mass and field renormalization appear together. In both cases the counterterms are fixed by the same exact pole requirements, whether the loop integrals are divergent or finite.

Vertex counterterms have also appeared, but only as much as is necessary to keep the perturbative theory complete. Their systematic treatment belongs later. At this stage the central lesson is narrower and more fundamental: a two-point function is renormalized correctly when its perturbative pole reproduces the physical one-particle state.

Assignment 1 asks the reader to use these ideas rather than merely repeat them. Problem 1 first derives Feynman’s general parametrization and then applies it in six-dimensional ϕ^3 theory, where the one-loop self-energy already requires both mass and field renormalization and the one-loop vertex is a genuine three-denominator triangle. Problem 2 separates ultraviolet renormalization of a composite operator from its one-particle LSZ pole weight. Problem 3 extracts the pole mass and pole residue of a fermion directly from the structure of its Yukawa self-energy. The three problems are deliberately different: the first develops a general loop technique, the second tests the operator viewpoint of Part I, and the third combines the same loop technology with spinor structure.

Assignment 1

The three problems below are calculations based on Sections 6.5–6.10. They are meant to test different pieces of the same logic rather than repeat the worked examples. Problem 1 uses a new theory and makes Feynman’s parametrization itself the subject of the calculation. Problem 2 separates ultraviolet renormalization of a composite insertion from its one-particle pole weight. Problem 3 treats a fermion self-energy with a non-trivial numerator. Use the dimension stated in each problem and define

$$\frac{1}{\bar{\epsilon}} \equiv \frac{1}{\epsilon} - \gamma_E + \ln 4\pi. \quad (326)$$

All loop-momentum master integrals needed below are supplied. Unless a part explicitly asks otherwise, it is enough to leave the remaining Feynman-parameter integrals unevaluated.

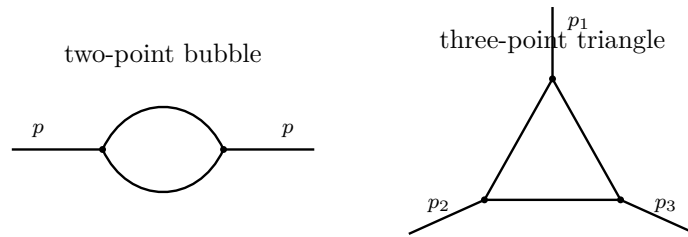
Problem 1: Feynman’s parametrization and renormalization in six-dimensional ϕ^3 theory

Consider the dimensionally regulated theory

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\mu_R^\epsilon g}{3!}\phi^3, \quad d = 6 - 2\epsilon. \quad (327)$$

The coupling g is dimensionless at six dimensions. A real cubic potential is not bounded below, so this model is being used here only as a perturbative laboratory around $\phi = 0$. A linear counterterm may be chosen so that the renormalized one-point function vanishes; that tadpole condition does not enter the one-particle-irreducible two- and three-point graphs studied below.

The two one-loop topologies needed in this problem are a bubble self-energy and a triangle vertex:



- (a) **Derive Feynman’s general formula.** After Wick rotation, start from the Schwinger representation

$$\frac{1}{A^\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty ds s^{\alpha-1} e^{-sA}, \quad \Re A > 0. \quad (328)$$

Introduce one Schwinger parameter s_i for each denominator in $1/(A_1^{\alpha_1} \cdots A_n^{\alpha_n})$, and make the change of variables

$$s_i = \rho x_i, \quad \rho = \sum_{i=1}^n s_i, \quad x_i \geq 0, \quad \sum_{i=1}^n x_i = 1. \quad (329)$$

Show that the measure can be written as

$$\prod_{i=1}^n ds_i = \rho^{n-1} d\rho \prod_{i=1}^n dx_i \delta\left(1 - \sum_{i=1}^n x_i\right), \quad (330)$$

perform the ρ integral, and derive Eq. (259). Then specialize your result to unit powers and recover both

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{[xA + (1-x)B]^2} \quad (331)$$

and

$$\frac{1}{ABC} = 2 \int_0^\infty dx dy dz \delta(1-x-y-z) \frac{1}{(xA+yB+zC)^3}. \quad (332)$$

Explain in one sentence why the delta function makes the apparent integration region $[0, \infty)^n$ a finite simplex.

(b) **Two denominators: the one-loop self-energy.** Use the convention

$$\Gamma^{(2)}(p^2) = p^2 - m^2 + \Pi(p^2), \quad (333)$$

so that the one-loop bubble obeys

$$i\Pi^{(1)}(p^2) = \frac{1}{2}(-i\mu_R^\epsilon g)^2 \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2 + i0} \frac{i}{(k-p)^2 - m^2 + i0}. \quad (334)$$

Apply the two-denominator formula, complete the square explicitly, and show that after shifting the loop momentum

$$\Delta_2(x, p^2) = m^2 - x(1-x)p^2. \quad (335)$$

You may then use the six-dimensional master integral

$$\mu_R^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i0)^2} = -\frac{i\Delta}{(4\pi)^3} \left[\frac{1}{\epsilon} + 1 - \ln \frac{\Delta - i0}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \quad (336)$$

Show that

$$\Pi^{(1)}(p^2) = -\frac{g^2}{2(4\pi)^3} \int_0^1 dx \Delta_2(x, p^2) \left[\frac{1}{\epsilon} + 1 - \ln \frac{\Delta_2(x, p^2) - i0}{\mu_R^2} \right] \quad (337)$$

and hence extract the ultraviolet polynomial

$$\Pi_{\text{div}}^{(1)}(p^2) = -\frac{g^2}{2(4\pi)^3} \left(m^2 - \frac{p^2}{6} \right) \frac{1}{\epsilon}. \quad (338)$$

Why does the presence of both a constant and a p^2 term require both mass and field counterterms already at one loop? With

$$\Pi_{\text{ct}}(p^2) = \delta Z_\phi(p^2 - m^2) - \delta m^2, \quad (339)$$

impose mass-shell conditions with $m = M$ and show

$$\delta m^2 = \Pi^{(1)}(M^2), \quad \delta Z_\phi = -\Pi^{(1)'}(M^2). \quad (340)$$

Write these counterterms as one-dimensional x integrals and verify that their divergent parts are

$$\delta Z_\phi|_{\text{div}} = -\frac{g^2}{12(4\pi)^3} \frac{1}{\epsilon}, \quad \delta m^2|_{\text{div}} = -\frac{5g^2 M^2}{12(4\pi)^3} \frac{1}{\epsilon}. \quad (341)$$

- (c) **Three denominators: the one-loop vertex.** Take all external momenta incoming, $p_1 + p_2 + p_3 = 0$, and choose the loop denominators

$$D_1 = k^2 - m^2 + i0, \quad D_2 = (k + p_1)^2 - m^2 + i0, \quad D_3 = (k - p_2)^2 - m^2 + i0. \quad (342)$$

The triangle contribution is

$$\mathcal{V}_{\text{loop}}^{(3)} = (-i\mu_R^\epsilon g)^3 \int \frac{d^d k}{(2\pi)^d} \frac{i}{D_1} \frac{i}{D_2} \frac{i}{D_3}. \quad (343)$$

Use Eq. (332), complete the square, and show that

$$\Delta_3(x, y, z) = m^2 - xy p_1^2 - xz p_2^2 - yz p_3^2, \quad x + y + z = 1. \quad (344)$$

You may use

$$\mu_R^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i0)^3} = -\frac{i}{2(4\pi)^3} \left[\frac{1}{\epsilon} - \ln \frac{\Delta - i0}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \quad (345)$$

to reduce the triangle to

$$\boxed{\mathcal{V}_{\text{loop}}^{(3)} = -\frac{i\mu_R^\epsilon g^3}{(4\pi)^3} \int_0^\infty dx dy dz \delta(1 - x - y - z) \left[\frac{1}{\epsilon} - \ln \frac{\Delta_3(x, y, z) - i0}{\mu_R^2} \right]}. \quad (346)$$

Use the volume of the simplex,

$$\int_0^\infty dx dy dz \delta(1 - x - y - z) = \frac{1}{2}, \quad (347)$$

to show that the ultraviolet divergence is independent of the external momenta. With a local vertex counterterm $-i\mu_R^\epsilon \delta g$, determine

$$\boxed{\delta g_{\text{MS}}^{(1)} = -\frac{g^3}{2(4\pi)^3} \frac{1}{\epsilon}}. \quad (348)$$

Finally suppose that the scalar has first been mass-shell normalized as in part (b). Define the coupling instead at the Euclidean symmetric three-point configuration. To avoid

confusing the Euclidean external vectors with the loop momentum k , call them q_i :

$$q_1 + q_2 + q_3 = 0, \quad q_i^2 = Q^2, \quad q_i \cdot q_j = -\frac{Q^2}{2} \quad (i \neq j), \quad (349)$$

so that $p_i^2 = -Q^2$ after continuation to Minkowski invariants. Show that

$$\Delta_{3,*} = m^2 + Q^2(xy + xz + yz), \quad (350)$$

and write, without performing the simplex integral, the full counterterm δg required by

$$\mathcal{V}_R^{(3)}|_{\mathcal{K}_*(Q)} = -i\mu_R^\epsilon g. \quad (351)$$

Which part of δg is forced by ultraviolet finiteness and which part is fixed by the chosen subtraction condition? Explain why, once the scalar field has been mass-shell normalized, this off-shell definition has no remaining ambiguity under the linear field redefinitions admitted in these lectures.

Problem 2: Compute the leading pole weight of a composite interpolator

Work with the dimensionally regulated perturbative theory

$$\mathcal{L}_{\text{reg}} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\mu_R^{2\epsilon}\lambda}{4!}\phi^4, \quad d = 4 - 2\epsilon, \quad (352)$$

and assume that the exact theory contains a stable scalar particle $|p\rangle$ of pole mass M . Define

$$C_\phi \equiv \langle \Omega | \phi(0) | p \rangle, \quad Z_\phi^{\text{pole}} = |C_\phi|^2. \quad (353)$$

The composite operator ϕ^3 can interpolate the same particle. Because ϕ^3 and $m^2\phi$ have different engineering dimensions away from four dimensions, it is convenient to use

$$\mathcal{O}_0(x) \equiv \mu_R^{2\epsilon}\phi^3(x), \quad (354)$$

which has the same d -dimensional mass dimension as $m^2\phi$. Your task is first to compute its one-particle overlap

$$C_3^{(0)} \equiv \langle \Omega | \mathcal{O}_0(0) | p \rangle, \quad (355)$$

then to renormalize the composite insertion, and finally to obtain its pole weight $Z_{[\phi^3]_R}^{\text{pole}} = |C_3^R|^2$.

- (a) Draw the leading diagram for $C_3^{(0)}$. One of the three fields at the composite insertion carries the external one-particle line; the remaining two contract into a tadpole. Count the three possible contractions and show that

$$C_3^{(0)} = 3C_\phi I(m) + \mathcal{O}(\lambda), \quad I(m) = \mu_R^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2 + i0}. \quad (356)$$

State clearly why the diagram computes the overlap $C_3^{(0)}$ rather than the pole weight itself, and how the latter is obtained from the overlap.

- (b) Use the dimensional-regularization result

$$I(m) = -\frac{m^2}{16\pi^2} \left[\frac{1}{\bar{\epsilon}} + 1 - \ln \frac{m^2}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \quad (357)$$

Is $C_3^{(0)}$ finite as $\epsilon \rightarrow 0$? Write its divergent part explicitly. This is a divergence of a *composite insertion*; it is not removed by the ordinary mass and field counterterms merely because the Lagrangian has already been renormalized.

- (c) For this one-particle matrix element, show that the divergence can be cancelled by mixing the composite insertion with $m^2\phi$,

$$[\phi^3]_R = \mu_R^{2\epsilon} \phi^3 + c_{31} m^2 \phi + \dots \quad (358)$$

The ellipsis allows the additional operators—including equation-of-motion operators—that may be required in a complete composite-operator basis; you are not asked to determine them here. Determine the $1/\bar{\epsilon}$ part of c_{31} in the $\overline{\text{MS}}$ scheme by demanding that

$$C_3^R \equiv \langle \Omega | [\phi^3]_R(0) | p \rangle \quad (359)$$

be finite. Then use Eq. (357) to obtain the finite leading expression for C_3^R and hence

$$Z_{[\phi^3]_R}^{\text{pole}} = |C_3^R|^2 \quad (360)$$

in terms of m , μ_R and Z_ϕ^{pole} . Explain why adding a finite multiple of $m^2\phi$ would change this pole weight while leaving the physical particle and every properly normalized S -matrix element unchanged.

- (d) Check the factor of three without referring to the Wick contraction. The regulated equation of motion is

$$(\square + m^2)\phi + \frac{\mu_R^{2\epsilon}\lambda}{3!}\phi^3 = 0. \quad (361)$$

Take its vacuum-to-one-particle matrix element and show that

$$C_3^{(0)} = \frac{6(M^2 - m^2)}{\lambda} C_\phi. \quad (362)$$

You may use the leading pole shift

$$M^2 - m^2 = \frac{\lambda}{2} I(m) + \mathcal{O}(\lambda^2) \quad (363)$$

to recover the result of part (a). Explain why this equation-of-motion identity checks the one-particle overlap but does not replace the ultraviolet definition of the renormalized composite operator.

Problem 3: Compute the one-loop fermion pole mass and pole residue in Yukawa theory

Consider

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi + \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}\mu^2\phi^2 - \mu_R^\epsilon g\phi\bar{\psi}\psi, \quad (364)$$

and assume that the fermion is a stable asymptotic particle. The complete renormalizable Yukawa theory contains the additional scalar interactions discussed in Section 6.9; they do not enter the fermion self-energy at order g^2 , so they are omitted from this exercise. Use the self-energy convention

$$S^{-1}(p) = \not{p} - m - \Sigma(\not{p}), \quad (365)$$

so that the one-particle-irreducible insertion is $-i\Sigma$. For the two-point counterterms use the convention

$$\mathcal{L}_{\text{ct}}^{(2)} = \delta Z_\psi \bar{\psi}(i\not{\partial} - m)\psi - \delta m \bar{\psi}\psi. \quad (366)$$

With this convention the δZ_ψ insertion vanishes between on-shell tree-level spinors at $\not{p} = m$, so the mass condition fixes δm directly and the residue condition fixes δZ_ψ .

(a) Draw the one-loop fermion self-energy and show that

$$-i\Sigma^{(1)}(\not{p}) = (-ig)^2 \mu_R^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{i(\not{p} - \not{k} + m)}{(p-k)^2 - m^2 + i0} \frac{i}{k^2 - \mu^2 + i0}. \quad (367)$$

Introduce one Feynman parameter, shift the loop momentum, and show that the denominator is governed by

$$\Delta(x, p^2) = xm^2 + (1-x)\mu^2 - x(1-x)p^2. \quad (368)$$

(b) You may use the standard integral

$$\mu_R^{2\epsilon} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i0)^2} = \frac{i}{16\pi^2} \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \quad (369)$$

With the sign convention stated above, your loop-momentum integration should reduce to

$$\Sigma^{(1)}(\not{p}) = -\frac{g^2}{16\pi^2} \int_0^1 dx [(1-x)\not{p} + m] \left[\frac{1}{\bar{\epsilon}} - \ln \frac{\Delta(x, p^2)}{\mu_R^2} \right] + \mathcal{O}(\epsilon). \quad (370)$$

Use this as a checkpoint, then write the result as

$$\Sigma^{(1)}(\not{p}) = \not{p} \Sigma_V^{(1)}(p^2) + m \Sigma_S^{(1)}(p^2). \quad (371)$$

Read off $\Sigma_V^{(1)}$ and $\Sigma_S^{(1)}$ as explicit integrals over x and determine their $1/\bar{\epsilon}$ parts. As a check, you should obtain

$$\Sigma_V^{(1)}|_{\text{div}} = -\frac{g^2}{32\pi^2} \frac{1}{\bar{\epsilon}}, \quad \Sigma_S^{(1)}|_{\text{div}} = -\frac{g^2}{16\pi^2} \frac{1}{\bar{\epsilon}}. \quad (372)$$

Is the one-loop self-energy finite before counterterms are included?

- (c) Let M denote the physical fermion pole mass. Starting from

$$[\not{p} - m - \Sigma(\not{p})]u(p) = 0, \quad p^2 = M^2, \quad (373)$$

derive

$$M [1 - \Sigma_V(M^2)] = m [1 + \Sigma_S(M^2)]. \quad (374)$$

Expand consistently to order g^2 and substitute your answer from part (b) to obtain $M - m$ as an explicit one-dimensional integral. Extract its ultraviolet pole. State which local counterterm has to cancel this divergence. In a mass-shell scheme, what additional finite condition is imposed on that counterterm?

- (d) Write the inverse propagator as

$$\Gamma_\psi^{(2)}(p) = A(p^2)\not{p} - B(p^2) \quad (375)$$

and show that

$$\boxed{(Z_\psi^{\text{pole}})^{-1} = A(M^2) + 2M^2 A'(M^2) - 2MB'(M^2)}. \quad (376)$$

Using $A = 1 - \Sigma_V$ and $B = m(1 + \Sigma_S)$ to one loop, substitute the x -integrals from part (b) and obtain an explicit one-dimensional expression for $(Z_\psi^{\text{pole}})^{-1}$. You may use

$$\frac{\partial}{\partial p^2} \ln \Delta(x, p^2) = -\frac{x(1-x)}{\Delta(x, p^2)}. \quad (377)$$

Which part is ultraviolet divergent? State what the kinetic counterterm δZ_ψ must cancel. In a mass-shell scheme, what finite residue condition is imposed in addition?

- (e) Finally consider $\psi\phi \rightarrow \psi\phi$. Take the external scalar field to have unit pole residue and let Γ_{amp} denote the connected Green function with all external full propagators removed. Write the LSZ relation between Γ_{amp} and the physical amplitude, including the two external fermion pole factors. Explain why this calculation uses only Z_ψ^{pole} and the one-particle projection of ψ , and does not require an operator identity $\psi = \sqrt{Z} \psi_R$.